

CE SAR Test Report

APPLICANT : Magne AI Global tech limited
EQUIPMENT : MAG1
BRAND NAME : MaQ
MODEL NAME : MA1
STANDARD : EN 50360:2017
EN 50360:2017/A1:2023
EN 50566:2017
EN 50566:2017/A1:2023
EN IEC 62311:2020
EN 50665:2017
EN 62479:2010
EN 50663:2017

We, Sporton International Inc. (Kunshan), would like to declare that the tested sample has been evaluated in accordance with the procedures and had been in compliance with the applicable technical standards.

The test results in this report apply exclusively to the tested model / sample. Without written approval of Sporton International Inc. (Kunshan), the test report shall not be reproduced except in full.



Approved by: Si Zhang

Sporton International Inc. (Kunshan)

**No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300
People's Republic of China**

Table of Contents

1. Statement of Compliance	4
2. Administration Data	5
3. Guidance Applied.....	6
4. Equipment Under Test (EUT)	7
4.1 General Information	7
5. Basic Restrictions.....	9
6. Specific Absorption Rate (SAR).....	10
6.1 Introduction	10
6.2 SAR Definition.....	10
7. System Description and Setup	11
7.1 E-Field Probe	12
7.2 Data Acquisition Electronics (DAE).....	12
7.3 Phantom.....	13
7.4 Device Holder.....	14
8. Measurement Procedures	15
8.1 Spatial Peak SAR Evaluation.....	15
8.2 Power Reference Measurement.....	16
8.3 Area & Zoom Scan Procedures.....	16
8.4 Volume Scan Procedures.....	17
8.5 SAR Averaged Methods.....	17
8.6 Power Drift Monitoring.....	17
9. Test Equipment List	18
10. System Verification	19
10.1 Tissue Simulating Liquids.....	19
10.2 Tissue Verification	20
10.3 System Performance Check Results.....	21
11. RF Exposure Positions.....	23
11.1 Ear and handset reference point	23
11.2 Definition of the cheek position.....	24
11.3 Definition of the tilt position.....	25
11.4 Body Device	25
11.5 Hand-Held Device	25
12. Conducted RF Output Power (Unit: dBm).....	26
13. Antenna Location	34
14. SAR Test Results	35
14.1 Head SAR	36
14.2 Body SAR	39
14.3 Limbs SAR	44
15. Simultaneous Transmission Analysis	49
15.1 Head Exposure Conditions	50
15.2 Body Exposure Conditions	51
15.3 Limbs Exposure Conditions	52
15.4 Multi-Band Combined SAR Result	53
16. Uncertainty Assessment	55
17. References.....	60
Appendixes.....	61
Appendix A. Plots of System Performance Check	61
Appendix B. Plots of High SAR Measurement	61
Appendix C. DASy Calibration Certificate.....	61
Appendix D. Test Setup Photos.....	61
Appendix E. Conducted RF Output Power Table	61

Revision History

REPORT NO.	VERSION	DESCRIPTION	ISSUED DATE
EA612311	Rev. 01	Initial issue of report	Jun. 02, 2026

1. Statement of Compliance

The maximum results of Specific Absorption Rate (SAR) found during testing for **Magne AI Global tech limited, MAG1, MA1**, are as follows.

Frequency Band		Highest SAR Summary		
		Head (Separation 0mm)	Body (Separation 5mm)	Limbs (Separation 0mm)
		10g SAR (W/kg)		
GSM	GSM900	0.182	0.760	2.256
	GSM1800	0.051	1.334	2.558
WCDMA	WCDMA I	0.099	1.463	3.115
	WCDMA V	0.134	0.310	1.277
	WCDMA VIII	0.112	0.437	1.213
LTE	LTE Band 1	0.045	1.300	2.667
	LTE Band 3	0.029	0.864	2.715
	LTE Band 5	0.100	0.689	1.238
	LTE Band 7	0.072	0.670	3.148
	LTE Band 8	0.081	0.915	1.009
	LTE Band 20	0.103	0.418	1.065
	LTE Band 26	0.092	1.137	1.601
	LTE Band 28	0.065	0.278	1.048
	LTE Band 40	0.056	1.230	2.641
	LTE Band 41/38	0.048	0.283	1.984
	LTE Band 42	0.280	0.305	0.587
NR	FR1 n1	0.053	1.482	3.133
	FR1 n3	0.051	1.437	3.102
	FR1 n5	0.122	0.477	1.391
	FR1 n7	0.097	0.711	2.859
	FR1 n8	0.126	0.570	1.317
	FR1 n20	0.147	0.480	1.389
	FR1 n28	0.137	0.322	0.882
	FR1 n40	0.115	1.454	2.945
	FR1 n41/n38	1.052	1.280	3.108
	FR1 n77/n78	0.682	1.242	2.825
WLAN	2.4GHz WLAN	0.831	0.700	1.664
	5GHz WLAN	0.120	0.264	0.730
Bluetooth	Bluetooth	0.045	0.061	0.091
NFC	NFC			0.011
Highest Simultaneous Transmission 10g SAR (W/kg)		1.883	1.980	3.939
Date of Testing:		2026/02/23 ~ 2026/06/01		

Remark:

1. This device supports LTE B38 and B41. Since the supported frequency span for LTE B38 falls completely within the support's frequency span for LTE B41, both LTE bands have the same target power, and both LTE bands share the same transmission path; therefore, SAR was only assessed for LTE B41.
2. This device supports 5GNR n38 / n78 and n41 / n77. Since the supported frequency span for 5GNR n38 / n78 falls completely within the support's frequency span for n41 / n77, both 5GNR bands have the same target power, and both 5GNR bands share the same transmission path; therefore, SAR was only assessed for n41 / n77.

Declaration of Conformity:
The test results with all measurement uncertainty excluded are presented in accordance with the regulation limits or requirements declared by manufacturers.
Comments and Explanations:
The declared of product specification for EUT presented in the report are provided by the manufacturer, and the manufacturer takes all the responsibilities for the accuracy of product specification.

According to REDCA Technical Guidance on the publication with a restriction in the Official Journal of Europe

of the reference of standard EN 50566: 2017, to ensuring the safe, a separation distance of 5 mm or less is applied for handheld device and body-mounted wireless communication devices used by the general public during SAR measurement. The 5mm distance is used for SAR testing for body, the 0mm distance is used for SAR testing for limbs, and this device is compliance with Specific Absorption Rate (SAR) for general population/uncontrolled exposure limits (Localized 10-gram SAR for head and trunk, limit: 2.0W/kg, Localized 10-gram SAR for Limbs, limit: 4.0W/kg) specified in Council Recommendation 1999/519/EC, and ICNIRP Guidelines, and RED (Directive 2014/53/EU), and Radio Equipment Regulations 2017, and had been tested in accordance with the measurement methods and procedures specified in EN IEC/IEEE 62209-1528:2021, EN 50360:2017, EN 50360:2017/A1:2023, EN50566:2017, EN 50566:2017/A1:2023, EN IEC 62311:2020, EN 50665:2017, EN 62479:2010, EN 50663:2017, EN 62209-1:2016 and EN 62209-2:2010, EN 62209-2:2010/A1:2019.

2. Administration Data

Sporton International Inc. (Kunshan) is accredited to ISO/IEC 17025:2017 by American Association for Laboratory Accreditation with Certificate Number 5145.02.

Testing Laboratory	
Test Firm	Sporton International Inc. (Kunshan)
Test Site Location	No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300 People's Republic of China TEL : +86-512-57900158
Test Site No.	SAR02-KS, SAR07-KS

Applicant	
Company Name	Magne AI Global tech limited
Address	FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

Manufacturer	
Company Name	Magne AI Global tech limited
Address	FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

3. Guidance Applied

The Specific Absorption Rate (SAR) testing specification, method, and procedure for this device is in accordance with the following standards.

- Council Recommendation 1999/519/EC
- EN 50360:2017
- EN 50360:2017/A1:2023
- EN 50566:2017
- EN 50566:2017/A1:2023
- EN IEC 62311:2020
- EN 62209-1:2016
- EN 62209-2:2010
- EN 62209-2:2010/A1:2019
- EN 50665:2017
- EN 62479:2010
- EN 50663:2017
- EN IEC/IEEE 62209-1528:2021

4. Equipment Under Test (EUT)

4.1 General Information

Product Feature & Specification	
Equipment Name	MAG1
Brand Name	MaQ
Model Name	MA1
IMEI Code	016813000001352/016813000001360, 016813000002657/016813000002665
Wireless Technology and Frequency Range	GSM900: 880 MHz ~ 915 MHz GSM1800: 1710 MHz ~ 1785 MHz WCDMA Band I: 1920 MHz ~ 1980 MHz WCDMA Band V: 824 MHz ~ 849 MHz WCDMA Band VIII: 880 MHz ~ 915 MHz LTE Band 1: 1920 MHz ~ 1980 MHz LTE Band 3: 1710 MHz ~ 1785 MHz LTE Band 5: 824 MHz ~ 849 MHz LTE Band 7: 2500 MHz ~ 2570 MHz LTE Band 8: 880 MHz ~ 915 MHz LTE Band 20: 832 MHz ~ 862 MHz LTE Band 26: 814 MHz ~ 849 MHz LTE Band 28: 703 MHz ~ 748 MHz LTE Band 38: 2570 MHz ~ 2620 MHz LTE Band 40: 2300 MHz ~ 2400 MHz LTE Band 41: 2496 MHz ~ 2690 MHz LTE Band 42: 3400 MHz ~ 3600 MHz 5G NR n1: 1920 MHz ~ 1980 MHz 5G NR n3: 1710 MHz ~ 1785 MHz 5G NR n5: 824 MHz ~ 849 MHz 5G NR n7: 2500 MHz ~ 2570 MHz 5G NR n8: 880 MHz ~ 915 MHz 5G NR n20: 832 MHz ~ 862 MHz 5G NR n28: 703 MHz ~ 748 MHz 5G NR n38: 2570 MHz ~ 2620 MHz 5G NR n40: 2300 MHz ~ 2400 MHz 5G NR n41: 2496 MHz ~ 2690 MHz 5G NR n77: 3300 MHz ~ 4200 MHz 5G NR n78: 3300 MHz ~ 3800 MHz 5G NR n79: 4400 MHz ~ 5000 MHz WLAN 2.4GHz Band: 2400 MHz ~ 2483.5 MHz WLAN 5GHz Band: 5150 MHz ~ 5350 MHz, 5470 MHz ~ 5725 MHz, 5725 MHz ~ 5850 MHz Bluetooth: 2400 MHz ~ 2483.5 MHz NFC: 13.56 MHz
Mode	GSM/GPRS RMC/AMR 12.2Kbps HSDPA HSUPA DC-HSDPA HSPA+ (16QAM uplink is supported) LTE: QPSK, 16QAM, 64QAM 5G NR: CP-OFDM / DFT-s-OFDM, PI/2 BPSK, QPSK, 16QAM, 64QAM, 256QAM WLAN 2.4GHz 802.11b/g/n HT20/HT40 WLAN 5GHz 802.11a/n HT20/HT40 WLAN 5GHz 802.11ac VHT20/VHT40/VHT80 Bluetooth BR/EDR/LE NFC: ASK
GSM / (E)GPRS Transfer mode	Class B – EUT cannot support Packet Switched and Circuit Switched Network simultaneously but can automatically switch between Packet and Circuit Switched Network.
EUT Stage	Production Unit
Remark: 1. This device supports VoIP in GPRS, WCDMA, LTE and 5GNR (e.g. for 3rd-party VoIP) and LTE supports VoLTE operation. 2. This device does not support DTM operation and supports GPRS mode up to multi-slot class 33.	

3. This device supports dual SIM dual standby. The WWAN radio transmission will be enabled by either one SIM at a time (single active).
4. The device implements receiver detect mechanism reduced power for SAR compliance at different exposure conditions (head, body, Limbs). The device will invoke corresponding work scenarios power level base on frequency bands/antennas, which can refer to appendix E. power table.
5. For 5G NR bands, using FTM to perform SAR with default 100% transmission.
6. This device supports HPUE mode for 5GNR n38/n41/n77/n78/n79 with higher power, so power class 2 was chosen to perform full SAR testing and power class 2 SAR can represent power class 3 SAR.
7. 5G NR NSA EN-DC mode, standalone SAR performed for 5GNR band with the maximum power, EN-DC SAR summed 5GNR standalone SAR and LTE standalone SAR, the result of EN-DC SAR is more conservatively.
8. This device supports 5GNR FR1 bands as following table, including NSA mode and SA mode.

<5G NR>

Mode	Band	Duplex	SCS(KHz)	Bandwidths(BW)
NSA	n77	TDD	30	10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100
	n78	TDD	30	10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100
	n79	TDD	30	40, 50, 60, 70, 80, 90, 100
SA	n1	FDD	15	5, 10, 15, 20, 25, 30, 40, 45, 50
	n3	FDD	15	5, 10, 15, 20, 25, 30, 35, 40, 45, 50
	n5	FDD	15	5, 10, 15, 20
	n7	FDD	15	5, 10, 15, 20, 25, 30, 35, 40, 50
	n8	FDD	15	5, 10, 15, 20
	n20	FDD	15	5, 10, 15, 20
	n28	FDD	15	5, 10, 15, 20, 25, 30
	n38	TDD	30	10, 15, 20, 25, 30, 40
	n40	TDD	30	10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100
	n41	TDD	30	10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100
	n77	TDD	30	10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100
	n78	TDD	30	10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100
	n79	TDD	30	40, 50, 60, 70, 80, 90, 100

5. Basic Restrictions

Depending on frequency, the following physical quantities (dosimetric/exposimetric quantities) are used to specify the basic restrictions on electromagnetic fields:

- between 0 and 1 Hz basic restrictions are provided for magnetic flux density for static magnetic fields (0 Hz) and current density for time-varying fields up to 1 Hz, in order to prevent effects on the cardiovascular and central nervous system.
- Between 1 Hz and 10 MHz basic restrictions are provided for current density to prevent effects on nervous system functions.
- Between 100 kHz and 10 GHz basic restrictions on SAR are provided to prevent whole-body heat stress and excessive localized heating of tissues. In the range 100 kHz to 10 MHz, restrictions on both current density and SAR are provided.
- Between 10 GHz and 300 GHz basic restrictions on power density are provided to prevent heating in tissue at or near the body surface.

The basic restrictions, given in Table 1, are set so as to account for uncertainties related to individual sensitivities, environmental conditions, and for the fact that the age and health status of members of the public vary.

Table 1

Basic restrictions for electric, magnetic and electromagnetic fields
(0 Hz to 300 GHz)

Frequency range	Magnetic flux density (mT)	Current density (mA/m ²) (rms)	Whole body average SAR (W/kg)	Localised SAR (head and trunk) (W/kg)	Localised SAR (limbs) (W/kg)	Power density, S (W/m ²)
0 Hz	40	—	—	—	—	—
>0-1 Hz	—	8	—	—	—	—
1-4 Hz	—	8/f	—	—	—	—
4-1 000 Hz	—	2	—	—	—	—
1 000 Hz-100 kHz	—	f/500	—	—	—	—
100 kHz-10 MHz	—	f/500	0,08	2	4	—
10 MHz-10 GHz	—	—	0,08	2	4	—
10-300 GHz	—	—	—	—	—	10

6. Specific Absorption Rate (SAR)

6.1 Introduction

SAR is related to the rate at which energy is absorbed per unit mass in an object exposed to a radio field. The SAR distribution in a biological body is complicated and is usually carried out by experimental techniques or numerical modeling. The standard recommends limits for two tiers of groups, occupational/controlled and general population/uncontrolled, based on a person's awareness and ability to exercise control over his or her exposure. In general, occupational/controlled exposure limits are higher than the limits for general population/uncontrolled.

6.2 SAR Definition

The SAR definition is the time derivative (rate) of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dv) of a given density (ρ). The equation description is as below:

$$SAR = \frac{d}{dt} \left(\frac{dW}{dm} \right) = \frac{d}{dt} \left(\frac{dW}{\rho dv} \right)$$

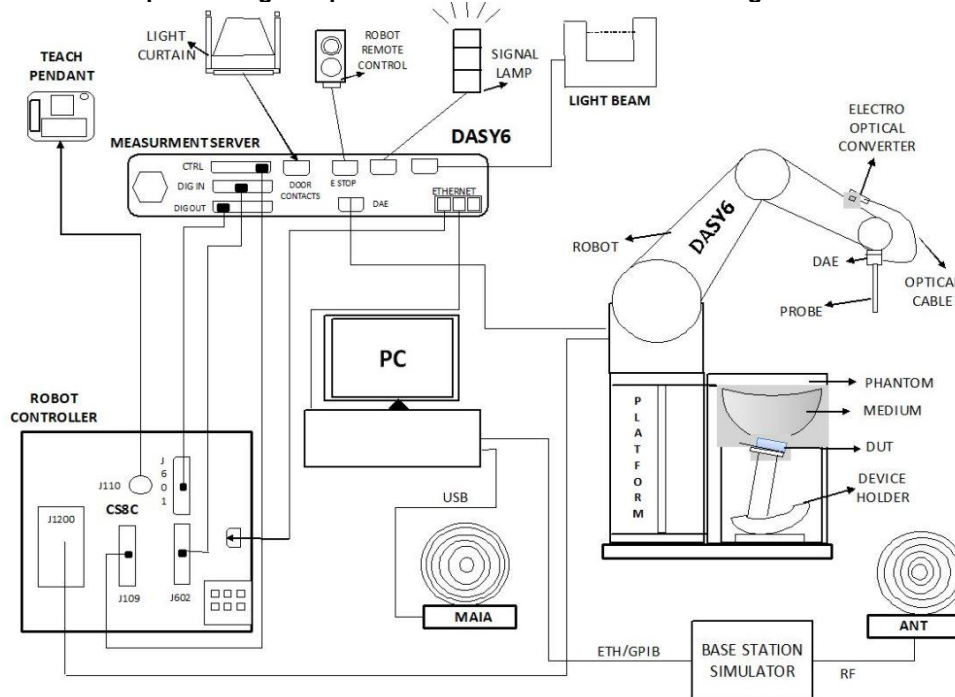
SAR is expressed in units of Watts per kilogram (W/kg)

$$SAR = \frac{\sigma |E|^2}{\rho}$$

Where: σ is the conductivity of the tissue, ρ is the mass density of the tissue and E is the RMS electrical field strength.

7. System Description and Setup

The DASY system used for performing compliance tests consists of the following items:



- A standard high precision 6-axis robot with controller, teach pendant and software. An arm extension for accommodating the data acquisition electronics (DAE).
- An isotropic Field probe optimized and calibrated for the targeted measurement.
- A data acquisition electronics (DAE) which performs the signal amplification, signal multiplexing, AD-conversion, offset measurements, mechanical surface detection, collision detection, etc. The unit is battery powered with standard or rechargeable batteries. The signal is optically transmitted to the EOC.
- The Electro-optical converter (EOC) performs the conversion from optical to electrical signals for the digital communication to the DAE. To use optical surface detection, a special version of the EOC is required. The EOC signal is transmitted to the measurement server.
- The function of the measurement server is to perform the time critical tasks such as signal filtering, control of the robot operation and fast movement interrupts.
- The Light Beam used is for probe alignment. This improves the (absolute) accuracy of the probe positioning.
- A computer running Win7 or Win10 and the DASY5 or DASY6 software.
- Remote control and teach pendant as well as additional circuitry for robot safety such as warning lamps, etc.
- The phantom, the device holder and other accessories according to the targeted measurement.

7.1 E-Field Probe

The SAR measurement is conducted with the dosimetric probe (manufactured by SPEAG). The probe is specially designed and calibrated for use in liquid with high permittivity. The dosimetric probe has special calibration in liquid at different frequency. This probe has a built in optical surface detection system to prevent from collision with phantom.

<ES3DV3 Probe>

Construction	Symmetric design with triangular core Interleaved sensors Built-in shielding against static charges PEEK enclosure material (resistant to organic solvents, e.g., DGBE)	
Frequency	10 MHz – 4 GHz; Linearity: ± 0.2 dB (30 MHz – 4 GHz)	
Directivity	± 0.2 dB in TSL (rotation around probe axis) ± 0.3 dB in TSL (rotation normal to probe axis)	
Dynamic Range	5 μ W/g – >100 mW/g; Linearity: ± 0.2 dB	
Dimensions	Overall length: 337 mm (tip: 20 mm) Tip diameter: 3.9 mm (body: 12 mm) Distance from probe tip to dipole centers: 2.0 mm	

<EX3DV4 Probe>

Construction	Symmetric design with triangular core Built-in shielding against static charges PEEK enclosure material (resistant to organic solvents, e.g., DGBE)	
Frequency	4 MHz – >10 GHz Linearity: ± 0.2 dB (30 MHz – 10 GHz)	
Directivity	± 0.3 dB in TSL (rotation around probe axis) ± 0.5 dB in TSL (rotation normal to probe axis)	
Dynamic Range	10 μ W/g – >100 mW/g Linearity: ± 0.2 dB (noise: typically <1 μ W/g)	
Dimensions	Overall length: 337 mm (tip: 20 mm) Tip diameter: 2.5 mm (body: 12 mm) Typical distance from probe tip to dipole centers: 1 mm	

7.2 Data Acquisition Electronics (DAE)

The data acquisition electronics (DAE) consists of a highly sensitive electrometer-grade preamplifier with auto-zeroing, a channel and gain-switching multiplexer, a fast 16 bit AD-converter and a command decoder and control logic unit. Transmission to the measurement server is accomplished through an optical downlink for data and status information as well as an optical uplink for commands and the clock.

The input impedance of the DAE is 200 MOhm; the inputs are symmetrical and floating. Common mode rejection is above 80 dB.



Photo of DAE

7.3 Phantom

<SAM Twin Phantom>

Shell Thickness	2 ± 0.2 mm; Center ear point: 6 ± 0.2 mm	
Filling Volume	Approx. 25 liters	
Dimensions	Length: 1000 mm; Width: 500 mm; Height: adjustable feet	
Measurement Areas	Left Hand, Right Hand, Flat Phantom	

The bottom plate contains three pair of bolts for locking the device holder. The device holder positions are adjusted to the standard measurement positions in the three sections. A white cover is provided to tap the phantom during off-periods to prevent water evaporation and changes in the liquid parameters. On the phantom top, three reference markers are provided to identify the phantom position with respect to the robot.

<ELI Phantom>

Shell Thickness	2 ± 0.2 mm (sagging: <1%)	
Filling Volume	Approx. 30 liters	
Dimensions	Major ellipse axis: 600 mm Minor axis: 400 mm	

The ELI phantom is intended for compliance testing of handheld and body-mounted wireless devices or for evaluating transmitters operating at low frequencies. ELI is fully compatible with standard and all known tissue simulating liquids.

7.4 Device Holder

<Mounting Device for Hand-Held Transmitter>

In combination with the Twin SAM V5.0/V5.0c or ELI phantoms, the Mounting Device for Hand-Held Transmitters enables rotation of the mounted transmitter device to specified spherical coordinates. At the heads, the rotation axis is at the ear opening. Transmitter devices can be easily and accurately positioned according to IEC 62209-1, IEEE 1528, FCC, or other specifications. The device holder can be locked for positioning at different phantom sections (left head, right head, flat). And upgrade kit to Mounting Device to enable easy mounting of wider devices like big smart-phones, e-books, small tablets, etc. It holds devices with width up to 140 mm.



Mounting Device for Hand-Held Transmitters



Mounting Device Adaptor for Wide-Phones

<Mounting Device for Laptops and other Body-Worn Transmitters>

The extension is lightweight and made of POM, acrylic glass and foam. It fits easily on the upper part of the mounting device in place of the phone positioned. The extension is fully compatible with the SAM Twin and ELI phantoms.



Mounting Device for Laptops

8. Measurement Procedures

The measurement procedures are as follows:

- (a) Use base station simulator (if applicable) or engineering software to transmit RF power continuously (continuous Tx) in the middle channel.
- (b) Keep EUT to radiate maximum output power or 100% duty factor (if applicable)
- (c) Measure output power through RF cable and power meter.
- (d) Place the EUT in the positions as setup photos demonstrates.
- (e) Set scan area, grid size and other setting on the DASY software.
- (f) Measure SAR transmitting at the middle channel for all applicable exposure positions.
- (g) Identify the exposure position and device configuration resulting the highest SAR
- (h) Measure SAR at the lowest and highest channels at the worst exposure position and device configuration.

According to the test standard, the recommended procedure for assessing the peak spatial-average SAR value consists of the following steps:

- (a) Power reference measurement
- (b) Area scan
- (c) Zoom scan
- (d) Power drift measurement

8.1 Spatial Peak SAR Evaluation

The procedure for spatial peak SAR evaluation has been implemented according to the test standard. It can be conducted for 1g and 10g, as well as for user-specific masses. The DASY software includes all numerical procedures necessary to evaluate the spatial peak SAR value.

The base for the evaluation is a "cube" measurement. The measured volume must include the 1g and 10g cubes with the highest averaged SAR values. For that purpose, the center of the measured volume is aligned to the interpolated peak SAR value of a previously performed area scan.

The entire evaluation of the spatial peak values is performed within the post-processing engine (SEMCAD). The system always gives the maximum values for the 1g and 10g cubes. The algorithm to find the cube with highest averaged SAR is divided into the following stages:

- (a) Extraction of the measured data (grid and values) from the Zoom Scan
- (b) Calculation of the SAR value at every measurement point based on all stored data (A/D values and measurement parameters)
- (c) Generation of a high-resolution mesh within the measured volume
- (d) Interpolation of all measured values from the measurement grid to the high-resolution grid
- (e) Extrapolation of the entire 3-D field distribution to the phantom surface over the distance from sensor to surface
- (f) Calculation of the averaged SAR within masses of 1g and 10g

8.2 Power Reference Measurement

The Power Reference Measurement and Power Drift Measurements are for monitoring the power drift of the device under test in the batch process. The minimum distance of probe sensors to surface determines the closest measurement point to phantom surface. This distance cannot be smaller than the distance of sensor calibration points to probe tip as defined in the probe properties.

8.3 Area & Zoom Scan Procedures

- a) Measure the local SAR at a test point within 8 mm of the phantom inner surface that is closest to the DUT.
- b) Measure the two-dimensional SAR distribution within the phantom (area scan procedure). The boundary of the measurement area shall not be closer than 20 mm from the phantom side walls. The distance between the measurement points should enable the detection of the location of local maximum with an accuracy of better than half the linear dimension of the tissue cube after interpolation. A maximum grid spacing of 20 mm for frequencies below 3 GHz and $(60/f \text{ [GHz]})$ mm for frequencies of 3 GHz and greater is recommended. The maximum distance between the geometrical center of the probe detectors and the inner surface of the phantom shall be 5 mm for frequencies below 3 GHz and $\ln(2)/2$ mm for frequencies of 3 GHz and greater, where $\ln(x)$ is the natural logarithm. The maximum variation of the sensor-phantom surface distance shall be 1 mm for frequencies below 3 GHz and 0,5 mm for frequencies of 3 GHz and greater. At all measurement points the angle of the probe with respect to the line normal to the surface should be less than 5°. If this cannot be achieved for a measurement distance to the phantom inner surface shorter than the probe diameter, additional uncertainty evaluation is needed.
- c) From the scanned SAR distribution, identify the position of the maximum SAR value, in addition identify the positions of any local maxima with SAR values within 2 dB of the maximum value that will not be within the zoom scan of other peaks; additional peaks shall be measured only when the primary peak is within 2 dB of the SAR compliance limit (e.g., 1 W/kg for 1,6 W/kg 1 g limit, or 1,26 W/kg for 2 W/kg, 10 g limit).
- d) Measure the three-dimensional SAR distribution at the local maxima locations identified in step c) (zoom scan procedure). The horizontal grid step shall be $(24 / f \text{ [GHz]})$ mm or less but not more than 8 mm. The minimum zoom scan size is 30 mm by 30 mm by 30 mm for frequencies below 3 GHz. For higher frequencies, the minimum zoom scan size can be reduced to 22 mm by 22 mm by 22 mm. The grid step in the vertical direction shall be $(8 \cdot f \text{ [GHz]})$ mm or less but not more than 5 mm, if uniform spacing is used (Annex C.3.3 of IEC 62209-1:2016). If variable spacing is used in the vertical direction, the maximum spacing between the two closest measured points to the phantom shell shall be $(12/f \text{ [GHz]})$ mm or less but not more than 4 mm, and the spacing between farther points shall increase by an incremental factor not exceeding 1,5. When variable spacing is used, extrapolation routines shall be tested with the same spacing as used in measurements. The maximum distance between the geometrical center of the probe detectors and the inner surface of the phantom shall be 5 mm for frequencies below 3 GHz and $\ln(2)/2$ mm for frequencies of 3 GHz and greater, where the plane wave skin depth and $\ln(x)$ is the natural logarithm. Separate grids shall be centered on each of the local SAR maxima found in step c). Uncertainties due to field distortion between the media boundary and the dielectric enclosure of the probe should also be minimized, which is achieved if the distance between the phantom surface and physical tip of the probe is larger than probe tip diameter. Other methods may utilize correction procedures for these boundary effects that enable high precision measurements closer than half the probe diameter. For all measurement points, the angle of the probe with respect to the flat phantom surface shall be less than 5°.

8.4 Volume Scan Procedures

The volume scan is used for assess overlapping SAR distributions for antennas transmitting in different frequency bands. It is equivalent to an oversized zoom scan used in standalone measurements. The measurement volume will be used to enclose all the simultaneous transmitting antennas. For antennas transmitting simultaneously in different frequency bands, the volume scan is measured separately in each frequency band. In order to sum correctly to compute the 1g or 10g aggregate SAR, the EUT remain in the same test position for all measurements and all volume scan use the same spatial resolution and grid spacing. When all volume scan were completed, the software, SEMCAD postprocessor can combine and subsequently superpose these measurement data to calculating the multiband SAR.

8.5 SAR Averaged Methods

In DASY, the interpolation and extrapolation are both based on the modified Quadratic Shepard's method. The interpolation scheme combines a least-square fitted function method and a weighted average method which are the two basic types of computational interpolation and approximation.

Extrapolation routines are used to obtain SAR values between the lowest measurement points and the inner phantom surface. The extrapolation distance is determined by the surface detection distance and the probe sensor offset. The uncertainty increases with the extrapolation distance. To keep the uncertainty within 1% for the 1 g and 10 g cubes, the extrapolation distance should not be larger than 5 mm.

8.6 Power Drift Monitoring

All SAR testing is under the EUT install full charged battery and transmit maximum output power. In DASY measurement software, the power reference measurement and power drift measurement procedures are used for monitoring the power drift of EUT during SAR test. Both these procedures measure the field at a specified reference position before and after the SAR testing. The software will calculate the field difference in dB. If the power drifts more than 5%, the SAR will be retested.

9. Test Equipment List

Manufacturer	Name of Equipment	Type/Model	Serial Number	Last Cal.	Due Date
SPEAG	750MHz System Validation Kit	D750V3	1087	Mar. 12, 2025	Mar. 11, 2026
SPEAG	835MHz System Validation Kit	D835V2	4d167	Dec. 02, 2025	Dec. 01, 2026
SPEAG	900MHz System Validation Kit	D900V2	1d186	Mar. 12, 2025	Mar. 11, 2026
SPEAG	1800MHz System Validation Kit	D1800V2	2d177	Nov. 22, 2023	Nov. 19, 2026
SPEAG	2000MHz System Validation Kit	D2000V2	1071	Mar. 13, 2025	Mar. 12, 2026
SPEAG	2300MHz System Validation Kit	D2300V2	1055	Aug. 21, 2023	Aug. 19, 2026
SPEAG	2450MHz System Validation Kit	D2450V2	1095	Feb. 08, 2024	Feb. 05, 2027
SPEAG	2600MHz System Validation Kit	D2600V2	1112	Dec. 18, 2023	Dec. 15, 2026
SPEAG	3300MHz System Validation Kit	D3300V2	1033	Jun. 13, 2025	Jun. 12, 2026
SPEAG	3500MHz System Validation Kit	D3500V2	1037	Nov. 20, 2023	Nov. 17, 2026
SPEAG	3700MHz System Validation Kit	D3700V2	1008	Nov. 20, 2023	Nov. 17, 2026
SPEAG	3900MHz System Validation Kit	D3900V2	1048	Mar. 09, 2023	Mar. 06, 2026
SPEAG	4200MHz System Validation Kit	D4200V2	1008	Aug. 13, 2025	Aug. 12, 2026
SPEAG	4600MHz System Validation Kit	D4600V2	1022	Aug. 13, 2025	Aug. 12, 2026
SPEAG	4900MHz System Validation Kit	D4900V2	1042	Mar. 13, 2023	Mar. 10, 2026
SPEAG	5000MHz System Validation Kit	D5GHzV2	1113	Sep. 17, 2025	Sep. 16, 2026
SPEAG	13MHz System Validation Kit	CLA13	1020	Jun. 05, 2025	Jun. 04, 2026
SPEAG	Data Acquisition Electronics	DAE4	1358	Jul. 15, 2025	Jul. 14, 2026
SPEAG	Data Acquisition Electronics	DAE4	1649	Aug. 27, 2025	Aug. 26, 2026
SPEAG	Data Acquisition Electronics	DAE4	1691	Jun. 04, 2025	Jun. 03, 2026
SPEAG	Dosimetric E-Field Probe	EX3DV4	7764	Oct. 22, 2025	Oct. 21, 2026
SPEAG	Dosimetric E-Field Probe	ES3DV3	3169	May 19, 2025	May 18, 2026
SPEAG	Dosimetric E-Field Probe	EX3DV4	7774	Jul. 25, 2025	Jul. 24, 2026
SPEAG	Dosimetric E-Field Probe	EX3DV4	7706	Feb. 24, 2026	Feb. 23, 2027
SPEAG	SAM Twin Phantom	SAM Twin	TP-1842	NCR	NCR
SPEAG	SAM Twin Phantom	SAM Twin	TP-2024	NCR	NCR
SPEAG	ELI Phantom	ELI V8.0	TP-2135	NCR	NCR
Beichuang	Thermo-Hygrometer	HTC-1	1959633	May 27, 2025	May 26, 2026
Beichuang	Thermo-Hygrometer	HTC-1	1959629	May 27, 2025	May 26, 2026
Beichuang	Thermo-Hygrometer	HTC-1	1959634	May 27, 2025	May 26, 2026
Beichuang	Thermo-Hygrometer	HTC-1	1980328	Apr. 11, 2026	Apr. 10, 2027
SPEAG	Phone Positioner	N/A	N/A	NCR	NCR
Anritsu	Radio Communication Analyzer	MT8820C	6201563900	Jul. 02, 2025	Jul. 01, 2026
Agilent	ENA Series Network Analyzer	E5071C	MY46112129	Jul. 02, 2025	Jul. 01, 2026
SPEAG	Dielectric Probe Kit	DAK-3.5	1144	Aug. 26, 2025	Aug. 25, 2026
SPEAG	Dielectric Probe Kit	DAK-12	1170	Sep. 22, 2025	Sep. 21, 2026
Anritsu	Vector Signal Generator	MG3710A	6201682672	Dec. 26, 2025	Dec. 25, 2026
Rohde & Schwarz	Vector Signal Generator	SMBV100A	175779	Dec. 24, 2025	Dec. 23, 2026
Rohde & Schwarz	Power Meter	NRVD	102081	Jul. 02, 2025	Jul. 01, 2026
Rohde & Schwarz	Power Sensor	NRV-Z5	100538	Jul. 02, 2025	Jul. 01, 2026
Rohde & Schwarz	Power Sensor	NRV-Z5	100539	Jul. 02, 2025	Jul. 01, 2026
Rohde & Schwarz	Power Sensor	NRP50S	101385	Oct. 14, 2025	Oct. 13, 2026
R&S	BLUETOOTH TESTER	CBT	101246	Jul. 03, 2025	Jul. 02, 2026
Rohde & Schwarz	Spectrum Analyzer	FSV7	101631	Oct. 11, 2025	Oct. 10, 2026
TES	DIGITAC THERMOMETER	TYPE-K	200505600	Jul. 05, 2025	Jul. 04, 2026
ARRA	Power Divider	A3200-2	N/A	Note 1	
MCL	Attenuation1	BW-S10W5+	N/A	Note 1	
MCL	Attenuation2	BW-S10W5+	N/A	Note 1	
MCL	Attenuation3	BW-S10W5+	N/A	Note 1	
BONN	POWER AMPLIFIER	BLMA 0830-3	087193A	Note 1	
BONN	POWER AMPLIFIER	BLMA 2060-2	087193B	Note 1	
BONN Elektronik	Power Amplifier	BSA 0125-150	076577A	Note 1	
Agilent	Dual Directional Coupler	778D	20500	Note 1	

Agilent	Dual Directional Coupler	11691D	MY48151020	Note 1
---------	--------------------------	--------	------------	--------

Note:

1. Prior to system verification and validation, the path loss from the signal generator to the system check source and the power meter, which includes the amplifier, cable, attenuator and directional coupler, was measured by the network analyzer. The reading of the power meter was offset by the path loss difference between the path to the power meter and the path to the system check source to monitor the actual power level fed to the system check source.
2. The dipole calibration interval can be extended to 3 years with justification. The dipoles are also not physically damaged, or repaired during the interval.
3. The justification data in appendix C can be found which the return loss is < -20dB, within 20% of prior calibration, the impedance is within 5 ohm of prior calibration for each dipole.

10. System Verification

10.1 Tissue Simulating Liquids

The tissue dielectric parameters of tissue-equivalent media used for SAR measurements must be characterized within a temperature range of 18°C to 25°C, measured with calibrated instruments and apparatuses, such as network analyzers and temperature probes. The temperature of the tissue-equivalent medium during SAR measurement must also be within 18°C to 25°C and within $\pm 2^\circ\text{C}$ of the temperature when the tissue parameters are characterized. For the measurement of the field distribution inside the SAM phantom with DASY, the phantom must be filled with around 25 liters of homogeneous body tissue simulating liquid. The liquid tissue depth was at least 15cm in the phantom for all SAR testing, which is shown in Fig. 12.1 and Fig.12.2.

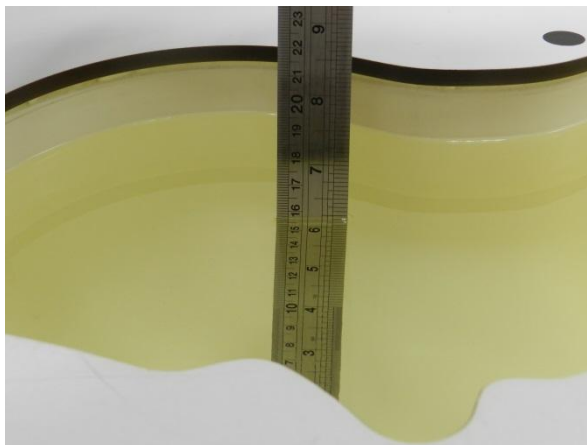


Fig 12.1 Photo of Liquid Height for Head SAR



Fig 12.2 Photo of Liquid Height for Body SAR

10.2 Tissue Verification

The following tissue formulations are provided for reference only as some of the parameters have not been thoroughly verified. The composition of ingredients may be modified accordingly to achieve the desired target tissue parameters required for routine SAR evaluation.

Frequency (MHz)	Water (%)	Sugar (%)	Cellulose (%)	Salt (%)	Preventol (%)	DGBE (%)	Conductivity (σ)	Permittivity (ϵ_r)
750	41.1	57.0	0.2	1.4	0.2	0	0.89	41.9
835	40.3	57.9	0.2	1.4	0.2	0	0.90	41.5
900	40.3	57.9	0.2	1.4	0.2	0	0.97	41.5
1800, 1900, 2000	55.2	0	0	0.3	0	44.5	1.40	40.0
2450	55.0	0	0	0	0	45.0	1.80	39.2
2600	54.8	0	0	0.1	0	45.1	1.96	39.0

Simulating Liquid for 5GHz, Manufactured by SPEAG

Ingredients	(% by weight)
Water	64~78%
Mineral oil	11~18%
Emulsifiers	9~15%
Additives and Salt	2~3%

<Tissue Dielectric Parameter Check Results>

Frequency (MHz)	Liquid Temp (°C)	Conductivity (σ)	Permittivity (ϵ_r)	Conductivity Target (σ)	Permittivity Target (ϵ_r)	Delta (σ) (%)	Delta (ϵ_r) (%)	Limit (%)	Date
750	22.8	0.89	42.94	0.89	41.9	0.00	2.48	±5.0	2026-02-23
750	22.9	0.886	43.024	0.89	41.9	-0.45	2.68	±5.0	2026-03-02
835	22.7	0.922	42.519	0.9	41.5	2.44	2.46	±5.0	2026-02-24
835	22.7	0.915	42.582	0.9	41.5	1.67	2.61	±5.0	2026-03-03
900	22.7	0.948	42.354	0.97	41.5	-2.27	2.06	±5.0	2026-02-25
900	22.8	0.95	42.343	0.97	41.5	-2.06	2.03	±5.0	2026-03-04
1800	22.6	1.412	40.555	1.4	40.0	0.86	1.39	±5.0	2026-02-26
1800	22.6	1.425	40.441	1.4	40.0	1.79	1.10	±5.0	2026-03-05
2000	22.9	1.439	39.237	1.4	40.0	2.79	-1.91	±5.0	2026-02-27
2000	22.6	1.428	39.194	1.4	40.0	2.00	-2.01	±5.0	2026-03-06
2300	22.6	1.608	39.522	1.67	39.5	-3.71	0.06	±5.0	2026-02-28
2300	22.7	1.612	39.534	1.67	39.5	-3.47	0.09	±5.0	2026-03-07
2450	22.8	1.761	39.329	1.8	39.2	-2.17	0.33	±5.0	2026-03-08
2600	22.7	1.897	38.784	1.96	39.0	-3.21	-0.55	±5.0	2026-03-01
2600	22.9	1.919	38.77	1.96	39.0	-2.09	-0.59	±5.0	2026-03-08
3300	22.6	2.77	37.873	2.7	38.13	2.59	-0.67	±5.0	2026-03-09
3500	22.8	2.984	37.772	2.91	37.9	2.54	-0.34	±5.0	2026-03-10
3700	22.6	3.181	37.611	3.12	37.7	1.96	-0.24	±5.0	2026-03-11
3900	22.9	3.409	37.316	3.33	37.51	2.37	-0.52	±5.0	2026-03-03
4100	22.8	3.578	37.017	3.53	37.28	1.36	-0.71	±5.0	2026-03-03
4400	22.9	3.9	37.1	3.84	36.92	1.56	0.49	±5.0	2026-03-08
4600	22.6	4.08	36.8	4.04	36.68	0.99	0.33	±5.0	2026-03-08
4800	22.7	4.26	36.7	4.25	36.43	0.24	0.74	±5.0	2026-03-08
4900	22.7	4.2	36.4	4.35	36.31	-3.45	0.25	±5.0	2026-03-08
4900	22.7	4.2	36.4	4.35	36.31	-3.45	0.25	±5.0	2026-03-08
5250	22.6	4.696	36.152	4.71	35.95	-0.30	0.56	±5.0	2026-03-14
5600	22.8	4.991	35.953	5.07	35.5	-1.56	1.28	±5.0	2026-03-14
5800	22.7	5.248	35.598	5.27	35.3	-0.42	0.84	±5.0	2026-03-14
13	22.7	0.744	56.2	0.75	55.00	-0.80	2.18	±5	2026-04-13
5800	22.8	5.205	35.757	5.27	35.3	-1.23	1.29	±5.0	2026-06-01

10.3 System Performance Check Results

Comparing to the original SAR value provided by SPEAG, the verification data should be within its specification of 10 %. Below table shows the target SAR and measured SAR after normalized to 1W input power. The table below indicates the system performance check can meet the variation criterion and the plots can be referred to Appendix A of this report.

Date	Frequency (MHz)	Input Power (mW)	Dipole S/N	Probe S/N	DAE S/N	Measured 1g SAR (W/kg)	Targeted 1g SAR (W/kg)	Normalized 1g SAR (W/kg)	Deviation (%)	Measured 10g SAR (W/kg)	Targeted 10g SAR (W/kg)	Normalized 10g SAR (W/kg)	Deviation (%)
2026-02-23	750	50.00	D750V3 - 1087	ES3DV3 - SN3169	DAE4 Sn1649	0.462	8.680	9.240	6.45	0.301	5.610	6.020	7.31
2026-03-02	750	50.00	D750V3 - 1087	ES3DV3 - SN3169	DAE4 Sn1649	0.461	8.680	9.220	6.22	0.303	5.610	6.060	8.02
2026-02-24	835	50.00	D835V2 - 4d167	ES3DV3 - SN3169	DAE4 Sn1649	0.459	9.360	9.180	-1.92	0.306	6.050	6.120	1.16
2026-03-03	835	50.00	D835V2 - 4d167	ES3DV3 - SN3169	DAE4 Sn1649	0.468	9.360	9.360	0.00	0.310	6.050	6.200	2.48
2026-02-25	900	50.00	D900V2 - 1d186	ES3DV3 - SN3169	DAE4 Sn1649	0.487	10.800	9.740	-9.81	0.322	6.930	6.440	-7.07
2026-03-04	900	50.00	D900V2 - 1d186	ES3DV3 - SN3169	DAE4 Sn1649	0.563	10.800	11.260	4.26	0.380	6.930	7.600	9.67
2026-02-26	1800	50.00	D1800V2 - 2d177	ES3DV3 - SN3169	DAE4 Sn1649	2.070	39.200	41.400	5.61	1.120	20.500	22.400	9.27
2026-03-05	1800	50.00	D1800V2 - 2d177	ES3DV3 - SN3169	DAE4 Sn1649	2.030	39.200	40.600	3.57	1.090	20.500	21.800	6.34
2026-02-27	2000	50.00	D2000V2 - 1071	ES3DV3 - SN3169	DAE4 Sn1649	2.170	41.000	43.400	5.85	1.140	21.300	22.800	7.04
2026-03-06	2000	50.00	D2000V2 - 1071	ES3DV3 - SN3169	DAE4 Sn1649	2.150	41.000	43.000	4.88	1.130	21.300	22.600	6.10
2026-02-28	2300	50.00	D2300V2 - 1055	ES3DV3 - SN3169	DAE4 Sn1649	2.470	48.400	49.400	2.07	1.220	23.700	24.400	2.95
2026-03-07	2300	50.00	D2300V2 - 1055	ES3DV3 - SN3169	DAE4 Sn1649	2.400	48.400	48.000	-0.83	1.140	23.700	22.800	-3.80
2026-03-08	2450	50.00	D2450V2 - 1095	EX3DV4 - SN7774	DAE4 Sn1649	2.640	52.600	52.800	0.38	1.260	24.700	25.200	2.02
2026-03-01	2600	50.00	D2600V2 - 1112	ES3DV3 - SN3169	DAE4 Sn1649	2.990	55.100	59.800	8.53	1.310	24.800	26.200	5.65
2026-03-08	2600	50.00	D2600V2 - 1112	ES3DV3 - SN3169	DAE4 Sn1649	2.750	55.100	55.000	-0.18	1.220	24.800	24.400	-1.61
2026-03-09	3300	50.00	D3300V2 - 1033	EX3DV4 - SN7774	DAE4 Sn1649	3.490	67.200	69.800	3.87	1.400	26.000	28.000	7.69
2026-03-10	3500	50.00	D3500V2 - 1037	EX3DV4 - SN7774	DAE4 Sn1649	3.270	65.400	65.400	0.00	1.280	24.700	25.600	3.64
2026-03-11	3700	50.00	D3700V2 - 1008	EX3DV4 - SN7774	DAE4 Sn1649	3.610	67.200	72.200	7.44	1.320	24.400	26.400	8.20
2026-03-03	3900	50.00	D3900V2 - 1048	EX3DV4 - SN7774	DAE4 Sn1649	3.640	69.100	72.800	5.35	1.290	24.100	25.800	7.05
2026-03-03	4100	50.00	D3900V2 - 1048	EX3DV4 - SN7774	DAE4 Sn1649	3.440	67.000	68.800	2.69	1.210	23.300	24.200	3.86
2026-03-08	4400	50.00	D4200V2 - 1008	EX3DV4 - SN7764	DAE4 Sn1358	3.600	67.500	72.000	6.67	1.250	23.000	25.000	8.70
2026-03-08	4600	50.00	D4600V2 - 1022	EX3DV4 - SN7764	DAE4 Sn1358	3.160	69.800	63.200	-9.46	1.080	23.000	21.600	-6.09
2026-03-08	4800	50.00	D4900V2 - 1042	EX3DV4 - SN7764	DAE4 Sn1358	3.630	69.200	72.600	4.91	1.170	21.900	23.400	6.85
2026-03-08	4900	50.00	D4900V2 - 1042	EX3DV4 - SN7764	DAE4 Sn1358	3.380	68.300	67.600	-1.02	1.070	21.700	21.400	-1.38
2026-03-08	4900	50.00	D4900V2 - 1042	EX3DV4 - SN7764	DAE4 Sn1358	3.580	68.300	71.600	4.83	1.110	21.700	22.200	2.30
2026-03-14	5250	50.00	D5GHzV2 - 1113	EX3DV4 - SN7774	DAE4 Sn1649	3.870	78.100	77.400	-0.90	1.120	22.400	22.400	0.00
2026-03-14	5600	50.00	D5GHzV2 - 1113	EX3DV4 - SN7774	DAE4 Sn1649	3.850	80.000	77.000	-3.75	1.090	23.000	21.800	-5.22
2026-03-14	5800	50.00	D5GHzV2 - 1113	EX3DV4 - SN7774	DAE4 Sn1649	3.710	77.900	74.200	-4.75	1.060	22.300	21.200	-4.93
2026-04-13	13	1000	CLA13 - SN:1020	EX3DV4 - SN7706	DAE4 Sn1691	0.527	0.548	0.527	-4.18	0.331	0.338	0.331	-2.65
2026-06-01	5800	50	D5GHzV2 - 1113	EX3DV4 - SN7774	DAE4 Sn1649	3.68	77.9	73.6	-5.52	1.03	22.3	20.6	-7.62

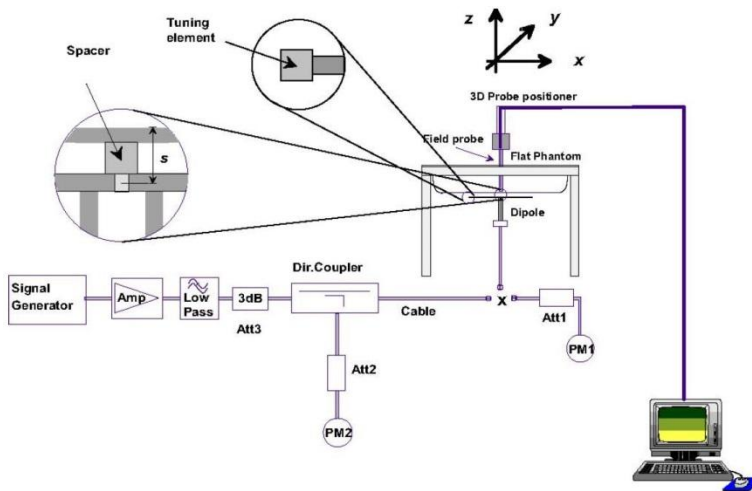


Fig 12.2.1 System Performance Check Setup

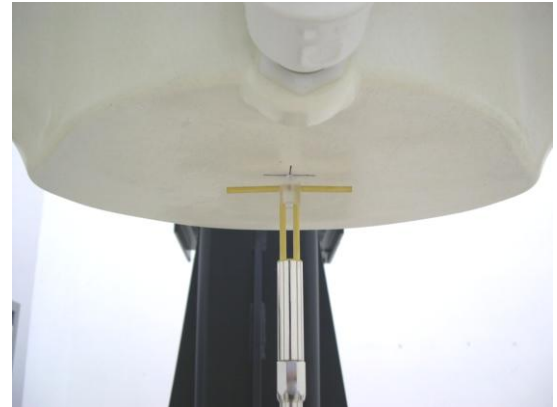


Fig 12.2.2 Setup Photo



Fig 12.2.3 Setup Photo

11. RF Exposure Positions

11.1 Ear and handset reference point

Figure 11.1.1 shows the front, back, and side views of the SAM phantom. The center-of-mouth reference point is labeled "M," the left ear reference point (ERP) is marked "LE," and the right ERP is marked "RE." Each ERP is 15 mm along the B-M (back-mouth) line behind the entrance-to-ear-canal (EEC) point, as shown in Figure 11.1.2 The Reference Plane is defined as passing through the two ear reference points and point M. The line N-F (neck-front), also called the reference pivoting line, is normal to the Reference Plane and perpendicular to both a line passing through RE and LE and the B-M line (see Figure 11.1.3). Both N-F and B-M lines should be marked on the exterior of the phantom shell to facilitate handset positioning. Posterior to the N-F line the ear shape is a flat surface with 6 mm thickness at each ERP, and forward of the N-F line the ear is truncated, as illustrated in Figure 11.1.2. The ear truncation is introduced to preclude the ear lobe from interfering with handset tilt, which could lead to unstable positioning at the cheek.

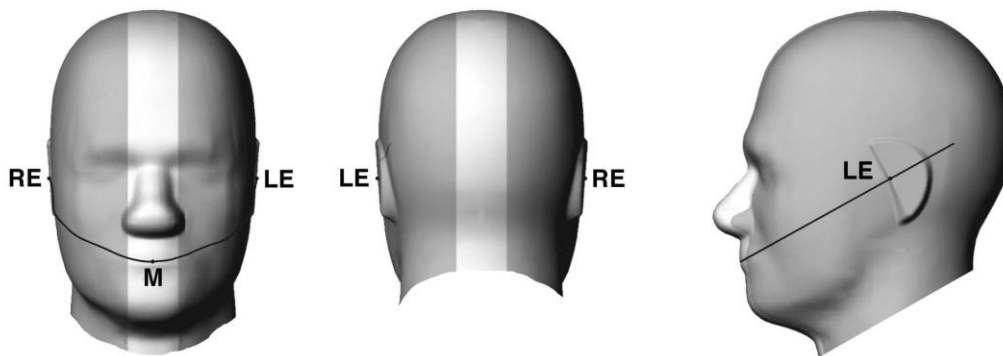


Fig 11.1.1 Front, back, and side views of SAM twin phantom

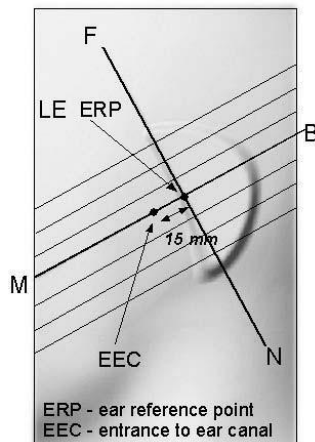


Fig 11.1.2 Close-up side view of phantom showing the ear region.

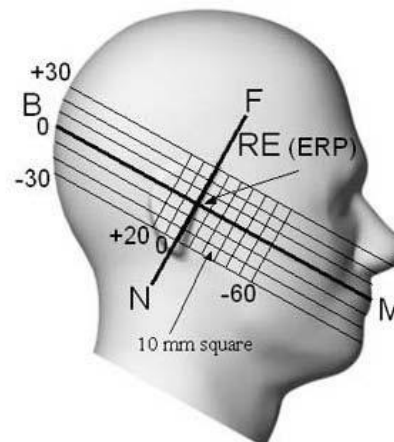


Fig 11.1.3 Side view of the phantom showing relevant markings and seven cross-sectional plane locations

11.2 Definition of the cheek position

1. Ready the handset for talk operation, if necessary. For example, for handsets with a cover piece (flip cover), open the cover. If the handset can transmit with the cover closed, both configurations must be tested.
2. Define two imaginary lines on the handset—the vertical centerline and the horizontal line. The vertical centerline passes through two points on the front side of the handset—the midpoint of the width w_t of the handset at the level of the acoustic output (point A in Figure 11.2.1 and Figure 11.2.2), and the midpoint of the width w_b of the bottom of the handset (point B). The horizontal line is perpendicular to the vertical centerline and passes through the center of the acoustic output (see Figure 11.2.1). The two lines intersect at point A. Note that for many handsets, point A coincides with the center of the acoustic output; however, the acoustic output may be located elsewhere on the horizontal line. Also note that the vertical centerline is not necessarily parallel to the front face of the handset (see Figure 11.2.2), especially for clamshell handsets, handsets with flip covers, and other irregularly-shaped handsets.
3. Position the handset close to the surface of the phantom such that point A is on the (virtual) extension of the line passing through points RE and LE on the phantom (see Figure 11.2.3), such that the plane defined by the vertical centerline and the horizontal line of the handset is approximately parallel to the sagittal plane of the phantom.
4. Translate the handset towards the phantom along the line passing through RE and LE until handset point A touches the pinna at the ERP.
5. While maintaining the handset in this plane, rotate it around the LE-RE line until the vertical centerline is in the plane normal to the plane containing B-M and N-F lines, i.e., the Reference Plane.
6. Rotate the handset around the vertical centerline until the handset (horizontal line) is parallel to the N-F line.
7. While maintaining the vertical centerline in the Reference Plane, keeping point A on the line passing through RE and LE, and maintaining the handset contact with the pinna, rotate the handset about the N-F line until any point on the handset is in contact with a phantom point below the pinna on the cheek. See Figure 11.2.3. The actual rotation angles should be documented in the test report.

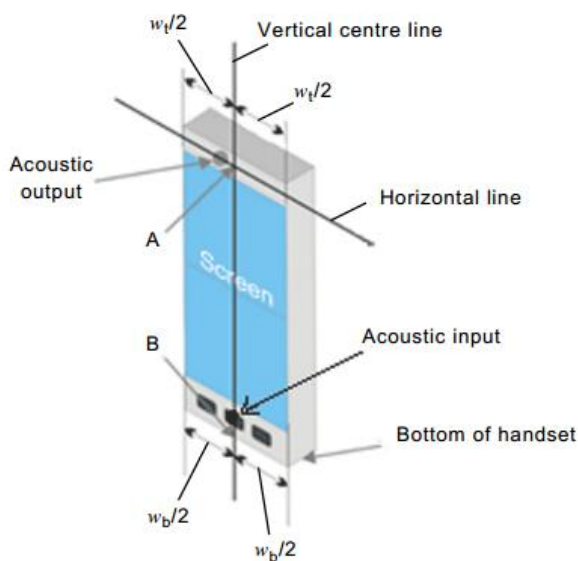


Fig 11.2.1 Handset vertical and horizontal reference lines—"a full touch screen smart phone"

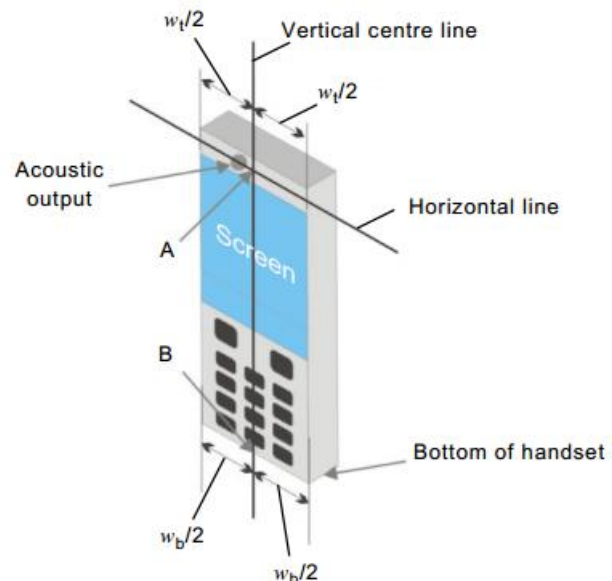


Fig 11.2.2 Handset vertical and horizontal reference lines—"a keyboard handset"



Fig 11.2.3 cheek or touch position. The reference points for the right ear (RE), left ear (LE), and mouth (M), which establish the Reference Plane for handset positioning, are indicated.

11.3 Definition of the tilt position

1. Ready the handset for talk operation, if necessary. For example, for handsets with a cover piece (flip cover), open the cover. If the handset can transmit with the cover closed, both configurations must be tested.
2. While maintaining the orientation of the handset, move the handset away from the pinna along the line passing through RE and LE far enough to allow a rotation of the handset away from the cheek by 15°.
3. Rotate the handset around the horizontal line by 15°.
4. While maintaining the orientation of the handset, move the handset towards the phantom on the line passing through RE and LE until any part of the handset touches the ear. The tilt position is obtained when the contact point is on the pinna. See below figure 11.3.1. If contact occurs at any location other than the pinna, e.g., the antenna at the back of the phantom head, the angle of the handset should be reduced. In this case, the tilt position is obtained if any point on the handset is in contact with the pinna and a second point

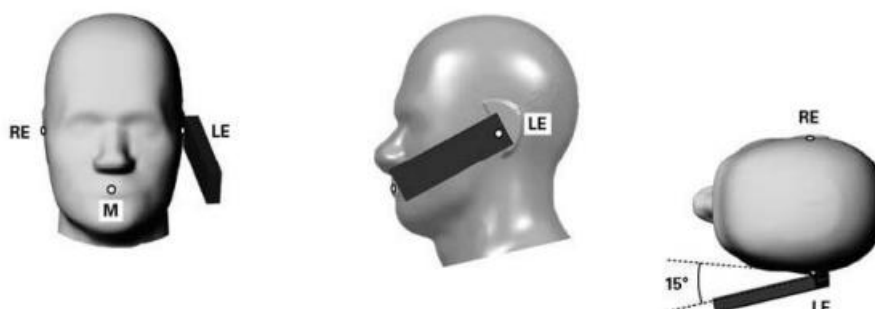


Fig 11.3.1 Tilt position. The reference points for the right ear (RE), left ear (LE), and mouth (M), which define the Reference Plane for handset positioning, are indicated.

11.4 Body Device

- (a) To position the device parallel to the phantom surface with all surfaces of the device.
- (b) To adjust the device parallel to the flat phantom.
- (c) To adjust the distance between the device surface and the flat phantom to 5mm.

11.5 Hand-Held Device

- (a) The device shall be placed directly against the flat phantom, for those sides of the device that are in contact with the hand during intended use.
- (b) To adjust the device parallel to the flat phantom.
- (c) To adjust the distance between the device surface and the flat phantom to 0mm.

12. Conducted RF Output Power (Unit: dBm)

The detailed conducted power table can refer to Appendix E.

<GSM Conducted Power>

General Note:

1. For SAR testing, according to EN62209 test procedure was based on the highest source-based time-average power.
2. Start tests at middle channel and determine the worst configuration for further high/low channel tests.

<WCDMA Conducted Power>

General Note:

Choose the RMC 12.2Kbps for each band at middle channel to test SAR and determine the worst configuration for further high/low channel test.

The following tests were conducted according to the test requirements outlines in 3GPP TS 34.121 specification.

A summary of these settings are illustrated below:

HSDPA Setup Configuration:

- a. The EUT was connected to Base Station Agilent E5515C referred to the Setup Configuration.
- b. The RF path losses were compensated into the measurements.
- c. A call was established between EUT and Base Station with following setting:
 - i. Set Gain Factors (β_c and β_d) and parameters were set according to each
 - ii. Specific sub-test in the following table, C10.1.4, quoted from the TS 34.121
 - iii. Set RMC 12.2Kbps + HSDPA mode.
 - iv. Set Cell Power = -86 dBm
 - v. Set HS-DSCH Configuration Type to FRC (H-set 1, QPSK)
 - vi. Select HSDPA Uplink Parameters
 - vii. Set Delta ACK, Delta NACK and Delta CQI = 8
 - viii. Set Ack-Nack Repetition Factor to 3
 - ix. Set CQI Feedback Cycle (k) to 4 ms
 - x. Set CQI Repetition Factor to 2
 - xi. Power Ctrl Mode = All Up bits
- d. The transmitted maximum output power was recorded.

Table C.10.1.4: β values for transmitter characteristics tests with HS-DPCCH

Sub-test	β_c	β_d	β_d (SF)	β_c/β_d	β_{hs} (Note 1, Note 2)	CM (dB) (Note 3)	MPR (dB) (Note 3)
1	2/15	15/15	64	2/15	4/15	0.0	0.0
2	12/15 (Note 4)	15/15 (Note 4)	64	12/15 (Note 4)	24/15	1.0	0.0
3	15/15	8/15	64	15/8	30/15	1.5	0.5
4	15/15	4/15	64	15/4	30/15	1.5	0.5
Note 1: Δ_{ACK}, Δ_{NACK} and $\Delta_{CQI} = 30/15$ with $\beta_{hs} = 30/15 * \beta_c$. Note 2: For the HS-DPCCH power mask requirement test in clause 5.2C, 5.7A, and the Error Vector Magnitude (EVM) with HS-DPCCH test in clause 5.13.1A, and HSDPA EVM with phase discontinuity in clause 5.13.1AA, Δ_{ACK} and $\Delta_{NACK} = 30/15$ with $\beta_{hs} = 30/15 * \beta_c$, and $\Delta_{CQI} = 24/15$ with $\beta_{hs} = 24/15 * \beta_c$. Note 3: CM = 1 for $\beta_c/\beta_d = 12/15$, $\beta_{hs}/\beta_c = 24/15$. For all other combinations of DPDCH, DPCCH and HS-DPCCH the MPR is based on the relative CM difference. This is applicable for only UEs that support HSDPA in release 6 and later releases. Note 4: For subtest 2 the β_c/β_d ratio of 12/15 for the TFC during the measurement period (TF1, TF0) is achieved by setting the signalled gain factors for the reference TFC (TF1, TF1) to $\beta_c = 11/15$ and $\beta_d = 15/15$.							

Setup Configuration

HSUPA Setup Configuration:

- The EUT was connected to Base Station Agilent E5515C referred to the Setup Configuration.
- The RF path losses were compensated into the measurements.
- A call was established between EUT and Base Station with following setting * :
 - Call Configs = 5.2B, 5.9B, 5.10B, and 5.13.2B with QPSK
 - Set the Gain Factors (β_c and β_d) and parameters (AG Index) were set according to each specific sub-test in the following table, C11.1.3, quoted from the TS 34.121
 - Set Cell Power = -86 dBm
 - Set Channel Type = 12.2k + HSPA
 - Set UE Target Power
 - Power Ctrl Mode= Alternating bits
 - Set and observe the E-TFCI
 - Confirm that E-TFCI is equal to the target E-TFCI of 75 for sub-test 1, and other subtest's E-TFCI
- The transmitted maximum output power was recorded.

Table C.11.1.3: β values for transmitter characteristics tests with HS-DPCCH and E-DCH

Sub-test	β_c	β_d	β_d (SF)	β_c/β_d	β_{HS} (Note 1)	β_{ec}	β_{ed} (Note 4) (Note 5)	β_{ed} (SF)	β_{ed} (Codes)	CM (dB) (Note 2)	MPR (dB) (Note 2) (Note 6)	AG Index (Note 5)	E-TFCI
1	11/15 (Note 3)	15/15 (Note 3)	64	11/15 (Note 3)	22/15	209/25	1309/225	4	1	1.0	0.0	20	75
2	6/15	15/15	64	6/15	12/15	12/15	94/75	4	1	3.0	2.0	12	67
3	15/15	9/15	64	15/9	30/15	30/15	β_{ed1} : 47/15 β_{ed2} : 47/15	4	2	2.0	1.0	15	92
4	2/15	15/15	64	2/15	4/15	2/15	56/75	4	1	3.0	2.0	17	71
5	15/15	0	-	-	5/15	5/15	47/15	4	1	1.0	0.0	12	67

Note 1: For sub-test 1 to 4, Δ_{ACK} , Δ_{NACK} and $\Delta_{CQI} = 30/15$ with $\beta_{hs} = 30/15 * \beta_c$. For sub-test 5, Δ_{ACK} , Δ_{NACK} and $\Delta_{CQI} = 5/15$ with $\beta_{hs} = 5/15 * \beta_c$.

Note 2: CM = 1 for $\beta_c/\beta_d = 12/15$, $\beta_{hs}/\beta_c = 24/15$. For all other combinations of DPDCH, DPCCH, HS-DPCCH, E-DPDCH and E-DPCCH the MPR is based on the relative CM difference.

Note 3: For subtest 1 the β_c/β_d ratio of 11/15 for the TFC during the measurement period (TF1, TF0) is achieved by setting the signalled gain factors for the reference TFC (TF1, TF1) to $\beta_c = 10/15$ and $\beta_d = 15/15$.

Note 4: In case of testing by UE using E-DPDCH Physical Layer category 1, Sub-test 3 is omitted according to TS25.306 Table 5.1g.

Note 5: β_{ed} can not be set directly; it is set by Absolute Grant Value.

Note 6: For subtests 2, 3 and 4, UE may perform E-DPDCH power scaling at max power which could results in slightly smaller MPR values.

Setup Configuration

DC-HSDPA 3GPP release 8 Setup Configuration:

- a. The EUT was connected to Base Station Agilent E5515C referred to the Setup Configuration below
- b. The RF path losses were compensated into the measurements.
- c. A call was established between EUT and Base Station with following setting:
 - i. Set RMC 12.2Kbps + HSDPA mode.
 - ii. Set Cell Power = -25 dBm
 - iii. Set HS-DSCH Configuration Type to FRC (H-set 12, QPSK)
 - iv. Select HSDPA Uplink Parameters
 - v. Set Gain Factors (β_c and β_d) and parameters were set according to each Specific sub-test in the following table, C10.1.4, quoted from the TS 34.121
 - a). Subtest 1: $\beta_c/\beta_d=2/15$
 - b). Subtest 2: $\beta_c/\beta_d=12/15$
 - c). Subtest 3: $\beta_c/\beta_d=15/8$
 - d). Subtest 4: $\beta_c/\beta_d=15/4$
 - vi. Set Delta ACK, Delta NACK and Delta CQI = 8
 - vii. Set Ack-Nack Repetition Factor to 3
 - viii. Set CQI Feedback Cycle (k) to 4 ms
 - ix. Set CQI Repetition Factor to 2
 - x. Power Ctrl Mode = All Up bits
- d. The transmitted maximum output power was recorded.

The following tests were conducted according to the test requirements outlines in 3GPP TS 34.121 specification. A summary of these settings are illustrated below:

C.8.1.12 Fixed Reference Channel Definition H-Set 12

Table C.8.1.12: Fixed Reference Channel H-Set 12

Parameter	Unit	Value
Nominal Avg. Inf. Bit Rate	kbps	60
Inter-TTI Distance	TTI's	1
Number of HARQ Processes	Processes	6
Information Bit Payload (N_{INF})	Bits	120
Number Code Blocks	Blocks	1
Binary Channel Bits Per TTI	Bits	960
Total Available SML's in UE	SML's	19200
Number of SML's per HARQ Proc.	SML's	3200
Coding Rate		0.15
Number of Physical Channel Codes	Codes	1
Modulation		QPSK
Note 1: The RMC is intended to be used for DC-HSDPA mode and both cells shall transmit with identical parameters as listed in the table.		
Note 2: Maximum number of transmission is limited to 1, i.e., retransmission is not allowed. The redundancy and constellation version 0 shall be used.		

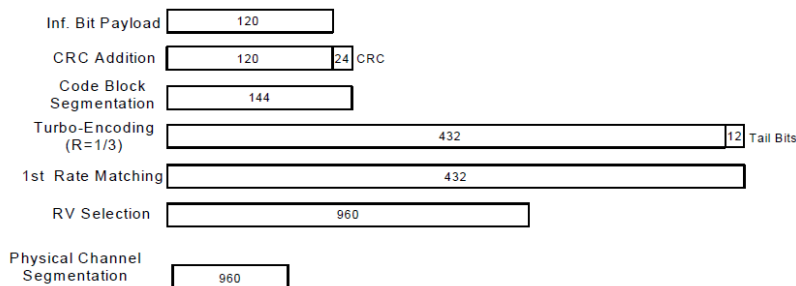


Figure C.8.19: Coding rate for Fixed reference Channel H-Set 12 (QPSK)

Setup Configuration

HSPA+ 3GPP release 7 (uplink category 7) 16QAM, Setup Configuration:

- a. The EUT was connected to Base Station Agilent E5515C referred to the Setup Configuration.
- b. The RF path losses were compensated into the measurements.
- c. A call was established between EUT and Base Station with following setting * :
 - i. Call Configs = 5.2E:HSPA+:UL with 16QAM
 - ii. Set the Gain Factors (β_c and β_d) and parameters (AG Index) were set according to each specific sub-test in the following table, C11.1.4, quoted from the TS 34.121-1 s5.2E
 - iii. Set Channel Parms
 - iv. Set Cell Power = -86 dBm
 - v. Set Channel Type = HSPA
 - vi. Set UE Target Power =21 dBm
 - vii. Power Ctrl Mode= All Up Bits
 - viii. Set Manual Uplink DPCH Bc/Bd = Manual
 - ix. Set Manual Uplink DPCH Bc and Bd=15,15(for 34.121-1 v8.10.0 table C11.1.4 sub-test 1)
 - x. Set HSPA Conn DL Channel Levels
 - xi. Set HS-SCCH Configs
 - xii. Set RB Test Mode Setup
 - xiii. Set Common HSUPA Parameters
 - xiv. Set Serving Grant
 - xv. Confirm that E-TFCI is equal to the target E-TFCI of 105 for sub-test 1, and other subtest's E-TFCI
- d. The transmitted maximum output power was recorded.

Table C.11.1.4: β values for transmitter characteristics tests with HS-DPCCH and E-DCH with 16QAM

Sub-test	β_c (Note3)	β_d	β_{HS} (Note1)	β_{ec}	β_{ed} (2xSF2) (Note 4)	β_{ed} (2xSF4) (Note 4)	CM (dB) (Note 2)	MPR (dB) (Note 2)	AG Index (Note 4)	E-TFCI (Note 5)	E-TFCI (boost)
1	1	0	30/15	30/15	β_{ed1} : 30/15 β_{ed2} : 30/15	β_{ed3} : 24/15 β_{ed4} : 24/15	3.5	2.5	14	105	105

Note 1: Δ_{ACK} , Δ_{NACK} and $\Delta_{CQI} = 30/15$ with $\beta_{HS} = 30/15 * \beta_c$.
 Note 2: CM = 3.5 and the MPR is based on the relative CM difference, MPR = MAX(CM-1,0).
 Note 3: DPDCH is not configured, therefore the β_c is set to 1 and $\beta_d = 0$ by default.
 Note 4: β_{ed} can not be set directly; it is set by Absolute Grant Value.
 Note 5: All the sub-tests require the UE to transmit 2SF2+2SF4 16QAM EDCH and they apply for UE using E-DPDCH category 7. E-DCH TTI is set to 2ms TTI and E-DCH table index = 2. To support these E-DCH configurations DPDCH is not allocated. The UE is signaled to use the extrapolation algorithm.

Setup Configuration

<FDD LTE Conducted Power>

General Note: Choose the highest output power among all configurations of bandwidths, RBs, modulations to test SAR at middle channel and determine the worst configuration for further high/low channel test.

<TDD LTE SAR Measurement>

TDD LTE configuration setup for SAR measurement

SAR was tested with a fixed periodic duty factor according to the highest transmission duty factor implemented for the device and supported by 3GPP.

- 3GPP TS 36.211 section 4.2 for Type 2 Frame Structure and Table 4.2-2 for uplink-downlink configurations
- "special subframe S" contains both uplink and downlink transmissions, it has been taken into consideration to determine the transmission duty factor according to the worst case uplink and downlink cyclic prefix requirements for UpPTS

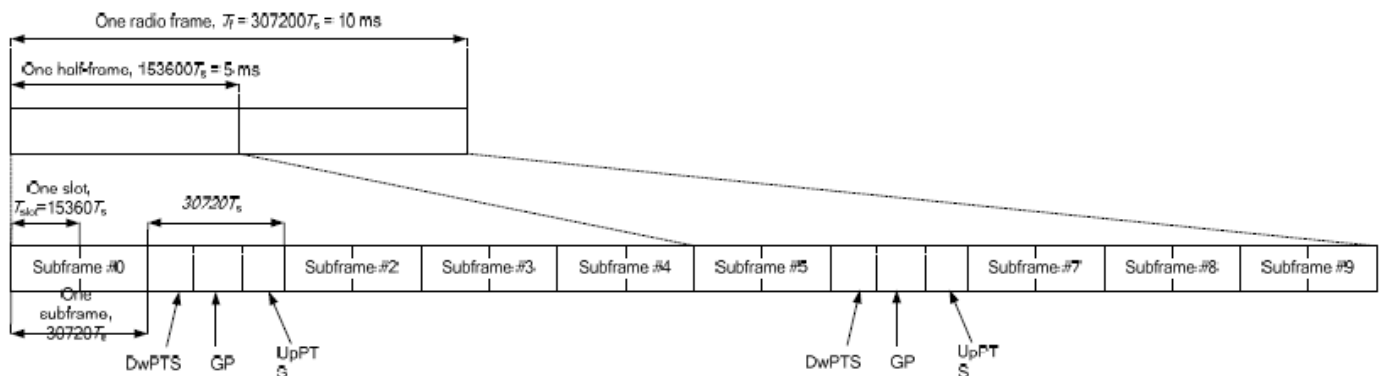


Figure 4.2-1: Frame structure type 2 (for 5 ms switch-point periodicity).

Table 4.2-2: Uplink-downlink configurations.

Uplink-downlink configuration	Downlink-to-Uplink Switch-point periodicity	Subframe number									
		0	1	2	3	4	5	6	7	8	9
0	5 ms	D	S	U	U	U	D	S	U	U	U
1	5 ms	D	S	U	U	D	D	S	U	U	D
2	5 ms	D	S	U	D	D	D	S	U	D	D
3	10 ms	D	S	U	U	U	D	D	D	D	D
4	10 ms	D	S	U	U	D	D	D	D	D	D
5	10 ms	D	S	U	D	D	D	D	D	D	D
6	5 ms	D	S	U	U	U	D	S	U	U	D

Table 4.2-1: Configuration of special subframe (lengths of DwPTS/GP/UpPTS).

Special subframe configuration	Normal cyclic prefix in downlink			Extended cyclic prefix in downlink		
	DwPTS	UpPTS		DwPTS	UpPTS	
		Normal cyclic prefix in uplink	Extended cyclic prefix in uplink		Normal cyclic prefix in uplink	Extended cyclic prefix in uplink
0	$6592 \cdot T_s$	$2192 \cdot T_s$	$2560 \cdot T_s$	$7680 \cdot T_s$	$2192 \cdot T_s$	$2560 \cdot T_s$
1	$19760 \cdot T_s$			$20480 \cdot T_s$		
2	$21952 \cdot T_s$			$23040 \cdot T_s$		
3	$24144 \cdot T_s$			$25600 \cdot T_s$		
4	$26336 \cdot T_s$			$7680 \cdot T_s$		
5	$6592 \cdot T_s$	$4384 \cdot T_s$	$5120 \cdot T_s$	$20480 \cdot T_s$	$4384 \cdot T_s$	$5120 \cdot T_s$
6	$19760 \cdot T_s$			$23040 \cdot T_s$		
7	$21952 \cdot T_s$			$12800 \cdot T_s$		
8	$24144 \cdot T_s$			-	-	-
9	$13168 \cdot T_s$			-	-	-

Special subframe (30720·T _s): Normal cyclic prefix in downlink (UpPTS)			
	Special subframe configuration	Normal cyclic prefix in uplink	Extended cyclic prefix in uplink
Uplink duty factor in one special subframe	0~4	7.13%	8.33%
	5~9	14.3%	16.7%

Special subframe(30720·T _s): Extended cyclic prefix in downlink (UpPTS)			
	Special subframe configuration	Normal cyclic prefix in uplink	Extended cyclic prefix in uplink
Uplink duty factor in one special subframe	0~3	7.13%	8.33%
	4~7	14.3%	16.7%

The highest duty factor is resulted from:

For LTE TDD Power class 3

- i. Uplink-downlink configuration: 0. In a half-frame consisted of 5 subframes, uplink operation is in 3 uplink subframes and 1 special subframe.
- ii. special subframe configuration: 5-9 for normal cyclic prefix in downlink, 4-7 for extended cyclic prefix in downlink
- iii. for special subframe with extended cyclic prefix in uplink, the total uplink duty factor in one half-frame is:
 $(3+0.167)/5 = 63.3\%$
- iv. for special subframe with normal cyclic prefix in uplink, the total uplink duty factor in one half-frame is:
 $(3+0.143)/5 = 62.9\%$
- v. For TDD LTE SAR measurement, the duty cycle 1:1.59 (62.9 %) was used perform testing and considering the theoretical duty cycle of 63.3% for extended cyclic prefix in the uplink, and the theoretical duty cycle of 62.9% for normal cyclic prefix in uplink, a scaling factor of extended cyclic prefix $63.3\%/62.9\% = 1.006$ is applied to scale-up the measured SAR result. The scaled TDD LTE SAR = measured SAR (W/kg)* Tune-up Scaling Factor* scaling factor for extended cyclic prefix.

<5G NR Conducted Power>

General Note:

1. 5G NR n1 / n3 / n5 / n7 / n8 / n20 / n28 / n38 / n40 / n41 / n77 / n78 / n79 support SA mode.
2. 5G NR n77 / n78 / n79 support NSA mode.
3. Choose the highest output power among all configurations of bandwidths, RBs, modulations to test SAR at middle channel and determine the worst configuration for further high/low channel test.
4. 5G NR operation does not have the fixed UL/DL frame structure, but during the transmitting/receiving it can be operated in the slot structure of 100% UL duty cycle, we are proposing the conservative way to evaluate SAR at 100% duty cycle. For the purpose of test 5G NR standalone SAR, and also test SAR level at 100% TX duty cycle.
5. For 5G NR bands, using FTM to perform SAR with default 100% transmission.
6. 5G NR supports CP-OFDM and DFT-s-OFDM modulation, for DFT-s-OFDM power is higher than CP-OFDM, so only show DFT-s-OFDM power table and chose DFT-s-OFDM to perform SAR testing.
7. For DFT-s-OFDM and CP-OFDM output power measurement reduction, according to 38.101 maximum power reduction for the CP-OFDM mode will not higher than DFT-s-OFDM mode, therefore, CP-OFDM measurement is unnecessary
8. NSA and SA mode should perform SAR separately. For the maximum power of SA mode is the same as NSA total power level, so SA standalone total power level SAR can represent NSA mode SAR.
9. 5G NR NSA mode, the power level is the same as 5G NR SA mode, so 5G NR NSA mode and SA mode power table only show one time.
10. 5G NR NSA EN-DC mode, standalone SAR performed for 5G NR band with the maximum power, EN-DC SAR summed 5G NR standalone SAR and LTE standalone SAR, the result of EN-DC SAR is more conservatively

<3GPP 38.101 MPR for EN-DC>

Table 6.2.2-1 Maximum power reduction (MPR) for power class 3

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	$\leq 3.5^1$	$\leq 1.2^1$	$\leq 0.2^1$
		$\leq 0.5^2$	$\leq 0.5^2$	0 ²
	QPSK	≤ 1		0
	16 QAM	≤ 2		≤ 1
	64 QAM		≤ 2.5	
CP-OFDM	256 QAM		≤ 4.5	
	QPSK	≤ 3		≤ 1.5
	16 QAM	≤ 3		≤ 2
	64 QAM		≤ 3.5	
			≤ 6.5	

NOTE 1: Applicable for UE operating in TDD mode with Pi/2 BPSK modulation and UE indicates support for UE capability powerBoosting-pi2BPSK and if the IE powerBoostPi2BPSK is set to 1 and 40 % or less slots in radio frame are used for UL transmission for bands n40, n41, n77, n78 and n79. The reference power of 0 dB MPR is 26 dBm.

NOTE 2: Applicable for UE operating in FDD mode, or in TDD mode in bands other than n40, n41, n77, n78 and n79 with Pi/2 BPSK modulation and if the IE powerBoostPi2BPSK is set to 0 and if more than 40 % of slots in radio frame are used for UL transmission for bands n40, n41, n77, n78 and n79.

Table 6.2.2-2 Maximum power reduction (MPR) for power class 2

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	≤ 3.5	≤ 0.5	0
	QPSK	≤ 3.5	≤ 1	0
	16 QAM	≤ 3.5	≤ 2	≤ 1
	64 QAM	≤ 3.5		≤ 2.5
	256 QAM		≤ 4.5	
CP-OFDM	QPSK	≤ 3.5	≤ 3	≤ 1.5
	16 QAM	≤ 3.5	≤ 3	≤ 2
	64 QAM		≤ 3.5	
	256 QAM		≤ 6.5	

Inter Band EN-DC Configuration

ENDC	Main Antenna Tx	
	LTE TX	NR TX
DC_1A_n77A	Ant 0	Ant 4
DC_1A_n78A	Ant 0	Ant 4
DC_1A_n79A	Ant 0	Ant 4
DC_3A_n77A	Ant 0	Ant 4
DC_3A_n78A	Ant 0	Ant 4
DC_3A_n79A	Ant 0	Ant 4
DC_41A_n77A	Ant 0	Ant 4
DC_41A_n78A	Ant 0	Ant 4

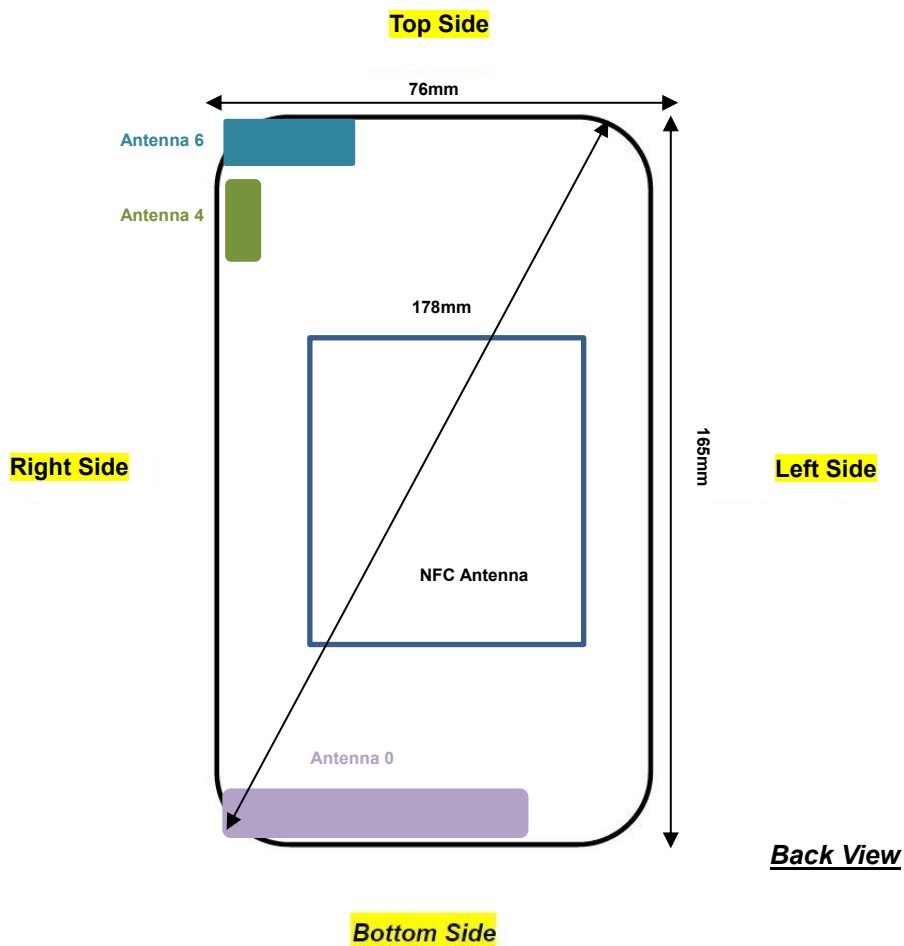
<Bluetooth Conducted Power>
General Note:

1. For Bluetooth SAR testing chose the mode with highest output power, which is 1Mbps at middle channel to test SAR and determine the worst configuration for further high/low channel testing.

<WLAN Conducted Power>

1. For 2.4GHz WLAN, choose the mode with highest output power, at middle channel to test SAR and determine the worst configuration for further high/low channel testing.
2. For 5.2&5.3&5.5&5.8GHz WLAN, choose the mode with highest output power, at middle channel to test SAR and determine the worst configuration for further high/low channel testing.

13. Antenna Location



Antenna	Support Band
Antenna 0	GSM: 900/1800<TX/RX> UMTS: B1/5/8<TX/RX> LTE: B1/3/5/7/8/20/26/28/38/40/41<TX/RX> 5G NR: n1/n3/n5/n7/n8/n20/n28/n40 <TX/RX>
Antenna 4	LTE: B42<TX/RX> 5G NR: n38/n41/n77/n78/n79<TX/RX>
Antenna 6	Bluetooth & WLAN 2.4GHz <TX/RX> WLAN 5GHz <TX/RX>
NFC Antenna	NFC <TX/RX>

14. SAR Test Results

General Note:

1. The reported SAR is the measured SAR value adjusted for maximum tune-up tolerance.
 - a. Tune-up scaling Factor = tune-up limit power (mW) / EUT RF power (mW), where tune-up limit is the maximum rated power among all production units.
 - b. For SAR testing of WLAN signal with non-100% duty cycle, the measured SAR is scaled-up by the duty cycle scaling factor which is equal to "1/(duty cycle)"
 - c. For SAR testing of Bluetooth signal with 83.3% theoretical duty cycle, the measured SAR is scaled-up by the duty cycle scaling factor which is equal to "1/(duty cycle) *83.3%".
 - d. For WWAN: Reported SAR(W/kg)= Measured SAR(W/kg)*Tune-up Scaling Factor
 - e. For WLAN/Bluetooth: Reported SAR(W/kg)= Measured SAR(W/kg)* Duty Cycle scaling factor * Tune-up scaling factor
 - f. For TDD LTE SAR measurement of power class 3, the duty cycle 1:1.59 (62.9 %) was used perform testing and considering the theoretical duty cycle of 63.3% for extended cyclic prefix in the uplink, and the theoretical duty cycle of 62.9% for normal cyclic prefix in uplink, a scaling factor of extended cyclic prefix 63.3%/62.9% = 1.006 is applied to scale-up the measured SAR result. The reported TDD LTE SAR (W/kg) = Measured SAR (W/kg)* Tune-up Scaling Factor* scaling factor for extended cyclic prefix.
2. When the WLAN transmission was verified using a spectrum analyzer.
3. The device implements receiver detect mechanism reduced power for SAR compliance at different exposure conditions (head, body, limbs). The device will invoke corresponding work scenarios power level base on frequency bands/antennas, which can refer to appendix E. power table.
4. For 5G NR bands, using FTM to perform SAR with default 100% transmission.
5. NFC mainly operate in hand-held extremity exposure conditions, therefore Standalone 10-g SAR testing for NFC will be performed with active mode and max power mode by test software with 100% duty cycle at 0mm separation distance.
6. NFC SAR test tissue-simulating liquid parameter: refer to EN IEC/IEEE 62209-1528:2021.
7. For SAR testing, the mode with the highest output power and the high-power channel were chosen to perform SAR measurement.
 - a. When the Reported SAR exceeds half of the SAR limit, choose the configuration for further tests on other channels.
 - b. When the Reported SAR in all frequency bands does not exceed half of the SAR limit, choose the worst-case configuration for further tests on other channels.

14.1 Head SAR

【GSM SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	GSM900	GPRS 2 Tx slots	Right Cheek	0mm	Ant 0	Receiver on	37	897.4	30.42	31.50	1.282	-0.04	0.137	0.176
	GSM900	GPRS 2 Tx slots	Right Tilted	0mm	Ant 0	Receiver on	37	897.4	30.42	31.50	1.282	-0.06	0.070	0.090
01	GSM900	GPRS 2 Tx slots	Left Cheek	0mm	Ant 0	Receiver on	37	897.4	30.42	31.50	1.282	0.08	0.142	0.182
	GSM900	GPRS 2 Tx slots	Left Tilted	0mm	Ant 0	Receiver on	37	897.4	30.42	31.50	1.282	0.03	0.079	0.101
02	GSM1800	GPRS 3 Tx slots	Right Cheek	0mm	Ant 0	Receiver on	698	1747.4	26.04	27.00	1.247	0.14	0.041	0.051
	GSM1800	GPRS 3 Tx slots	Right Tilted	0mm	Ant 0	Receiver on	698	1747.4	26.04	27.00	1.247	0.10	0.001	0.001
	GSM1800	GPRS 3 Tx slots	Left Cheek	0mm	Ant 0	Receiver on	698	1747.4	26.04	27.00	1.247	0.15	0.001	0.001
	GSM1800	GPRS 3 Tx slots	Left Tilted	0mm	Ant 0	Receiver on	698	1747.4	26.04	27.00	1.247	0.10	0.001	0.001

【WCDMA SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
03	WCDMA I	RMC 12.2Kbps	Right Cheek	0mm	Ant 0	Receiver on	9750	1950	22.42	24.00	1.439	0.05	0.069	0.099
	WCDMA I	RMC 12.2Kbps	Right Tilted	0mm	Ant 0	Receiver on	9750	1950	22.42	24.00	1.439	-0.09	0.033	0.047
	WCDMA I	RMC 12.2Kbps	Left Cheek	0mm	Ant 0	Receiver on	9750	1950	22.42	24.00	1.439	0.19	0.047	0.068
	WCDMA I	RMC 12.2Kbps	Left Tilted	0mm	Ant 0	Receiver on	9750	1950	22.42	24.00	1.439	0.13	0.053	0.076
	WCDMA V	RMC 12.2Kbps	Right Cheek	0mm	Ant 0	Receiver on	4182	836.4	22.15	23.00	1.216	-0.01	0.095	0.116
	WCDMA V	RMC 12.2Kbps	Right Tilted	0mm	Ant 0	Receiver on	4182	836.4	22.15	23.00	1.216	0.10	0.068	0.083
04	WCDMA V	RMC 12.2Kbps	Left Cheek	0mm	Ant 0	Receiver on	4182	836.4	22.15	23.00	1.216	0.04	0.110	0.134
	WCDMA V	RMC 12.2Kbps	Left Tilted	0mm	Ant 0	Receiver on	4182	836.4	22.15	23.00	1.216	-0.03	0.063	0.077
	WCDMA VIII	RMC 12.2Kbps	Right Cheek	0mm	Ant 0	Receiver on	2788	897.6	22.47	24.00	1.422	0.11	0.068	0.097
	WCDMA VIII	RMC 12.2Kbps	Right Tilted	0mm	Ant 0	Receiver on	2788	897.6	22.47	24.00	1.422	0.10	0.042	0.060
05	WCDMA VIII	RMC 12.2Kbps	Left Cheek	0mm	Ant 0	Receiver on	2788	897.6	22.47	24.00	1.422	0.06	0.079	0.112
	WCDMA VIII	RMC 12.2Kbps	Left Tilted	0mm	Ant 0	Receiver on	2788	897.6	22.47	24.00	1.422	0.11	0.044	0.063

【LTE SAR】

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
06	LTE Band 1	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	18300	1950	22.77	24.00	1.327			-0.17	0.034	0.045
	LTE Band 1	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	18300	1950	22.77	24.00	1.327			-0.11	0.020	0.027
	LTE Band 1	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	18300	1950	22.77	24.00	1.327			0.05	0.028	0.037
	LTE Band 1	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	18300	1950	22.77	24.00	1.327			-0.17	0.030	0.040
07	LTE Band 3	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	19575	1747.5	23.04	24.00	1.247			0.06	0.023	0.029
	LTE Band 3	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	19575	1747.5	23.04	24.00	1.247			0.04	0.001	0.001
	LTE Band 3	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	19575	1747.5	23.04	24.00	1.247			0.13	0.016	0.020
	LTE Band 3	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	19575	1747.5	23.04	24.00	1.247			-0.00	0.001	0.001
	LTE Band 5	10M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	20525	836.5	23.20	24.00	1.202			0.17	0.073	0.088
	LTE Band 5	10M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	20525	836.5	23.20	24.00	1.202			-0.04	0.045	0.054
08	LTE Band 5	10M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	20525	836.5	23.20	24.00	1.202			0.04	0.083	0.100
	LTE Band 5	10M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	20525	836.5	23.20	24.00	1.202			0.08	0.048	0.058
	LTE Band 7	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	21100	2535	23.11	24.00	1.227			-0.16	0.026	0.032
	LTE Band 7	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	21100	2535	23.11	24.00	1.227			0.01	0.021	0.026
09	LTE Band 7	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	21100	2535	23.11	24.00	1.227			0.08	0.059	0.072
	LTE Band 7	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	21100	2535	23.11	24.00	1.227			-0.09	0.033	0.040
	LTE Band 8	10M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	21625	897.5	22.91	24.00	1.285			0.14	0.055	0.071
	LTE Band 8	10M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	21625	897.5	22.91	24.00	1.285			0.12	0.001	0.001
10	LTE Band 8	10M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	21625	897.5	22.91	24.00	1.285			0.02	0.063	0.081
	LTE Band 8	10M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	21625	897.5	22.91	24.00	1.285			0.11	0.038	0.049
	LTE Band 20	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	24300	847	23.21	24.00	1.199			-0.02	0.081	0.097
	LTE Band 20	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	24300	847	23.21	24.00	1.199			0.18	0.053	0.064
11	LTE Band 20	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	24300	847	23.21	24.00	1.199			0.02	0.086	0.103
	LTE Band 20	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	24300	847	23.21	24.00	1.199			-0.05	0.055	0.066
	LTE Band 26	15M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	26865	831.5	22.82	24.00	1.312			0.04	0.063	0.083
	LTE Band 26	15M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	26865	831.5	22.82	24.00	1.312			-0.06	0.039	0.051
12	LTE Band 26	15M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	26865	831.5	22.82	24.00	1.312			0.08	0.070	0.092
	LTE Band 26	15M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	26865	831.5	22.82	24.00	1.312			0.11	0.043	0.056
13	LTE Band 28	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	27460	728	23.14	24.00	1.219			0.01	0.053	0.065
	LTE Band 28	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	27460	728	23.14	24.00	1.219			-0.04	0.035	0.043



CE SAR Test Report

Report No. : EA612311

	LTE Band 28	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	27460	728	23.14	24.00	1.219			0.00	0.051	0.062
	LTE Band 28	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	27460	728	23.14	24.00	1.219			0.06	0.034	0.041
	LTE Band 40	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	39150	2350	23.01	24.00	1.256	62.9	1.006	-0.02	0.025	0.032
	LTE Band 40	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	39150	2350	23.01	24.00	1.256	62.9	1.006	0.04	0.019	0.024
14	LTE Band 40	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	39150	2350	23.01	24.00	1.256	62.9	1.006	0.01	0.044	0.056
	LTE Band 40	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	39150	2350	23.01	24.00	1.256	62.9	1.006	0.18	0.028	0.035
	LTE Band 41	20M	QPSK	1	0	Right Cheek	0mm	Ant 0	Receiver on	40620	2593	23.01	24.00	1.256	62.9	1.006	0.08	0.017	0.021
	LTE Band 41	20M	QPSK	1	0	Right Tilted	0mm	Ant 0	Receiver on	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.06	0.001	0.001
15	LTE Band 41	20M	QPSK	1	0	Left Cheek	0mm	Ant 0	Receiver on	40620	2593	23.01	24.00	1.256	62.9	1.006	0.01	0.038	0.048
	LTE Band 41	20M	QPSK	1	0	Left Tilted	0mm	Ant 0	Receiver on	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.14	0.001	0.001
	LTE Band 42	20M	QPSK	1	0	Right Cheek	0mm	Ant 4	Receiver on	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.00	0.068	0.089
	LTE Band 42	20M	QPSK	1	0	Right Tilted	0mm	Ant 4	Receiver on	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.14	0.057	0.074
16	LTE Band 42	20M	QPSK	1	0	Left Cheek	0mm	Ant 4	Receiver on	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.02	0.215	0.280
	LTE Band 42	20M	QPSK	1	0	Left Tilted	0mm	Ant 4	Receiver on	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.05	0.141	0.184

【FR1 SAR】

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	FR1 n1	50M	BPSK	135	67	Right Cheek	0mm	Ant 0	Receiver on	390000	1950	22.96	24.00	1.271	0.09	0.030	0.038
	FR1 n1	50M	BPSK	135	67	Right Tilted	0mm	Ant 0	Receiver on	390000	1950	22.96	24.00	1.271	-0.14	0.018	0.023
17	FR1 n1	50M	BPSK	135	67	Left Cheek	0mm	Ant 0	Receiver on	390000	1950	22.96	24.00	1.271	0.05	0.042	0.053
	FR1 n1	50M	BPSK	135	67	Left Tilted	0mm	Ant 0	Receiver on	390000	1950	22.96	24.00	1.271	0.08	0.026	0.033
	FR1 n3	50M	BPSK	135	67	Right Cheek	0mm	Ant 0	Receiver on	349500	1747.5	22.82	24.00	1.312	-0.12	0.037	0.049
	FR1 n3	50M	BPSK	135	67	Right Tilted	0mm	Ant 0	Receiver on	349500	1747.5	22.82	24.00	1.312	0.04	0.001	0.001
18	FR1 n3	50M	BPSK	135	67	Left Cheek	0mm	Ant 0	Receiver on	349500	1747.5	22.82	24.00	1.312	0.07	0.039	0.051
	FR1 n3	50M	BPSK	135	67	Left Tilted	0mm	Ant 0	Receiver on	349500	1747.5	22.82	24.00	1.312	0.15	0.001	0.001
	FR1 n5	20M	BPSK	50	25	Right Cheek	0mm	Ant 0	Receiver on	167300	836.5	23.25	24.00	1.189	-0.10	0.099	0.118
	FR1 n5	20M	BPSK	50	25	Right Tilted	0mm	Ant 0	Receiver on	167300	836.5	23.25	24.00	1.189	0.02	0.059	0.070
19	FR1 n5	20M	BPSK	50	25	Left Cheek	0mm	Ant 0	Receiver on	167300	836.5	23.25	24.00	1.189	0.09	0.103	0.122
	FR1 n5	20M	BPSK	50	25	Left Tilted	0mm	Ant 0	Receiver on	167300	836.5	23.25	24.00	1.189	-0.07	0.054	0.064
	FR1 n7	50M	BPSK	135	67	Right Cheek	0mm	Ant 0	Receiver on	507000	2535	22.71	24.00	1.346	0.12	0.041	0.055
	FR1 n7	50M	BPSK	135	67	Right Tilted	0mm	Ant 0	Receiver on	507000	2535	22.71	24.00	1.346	0.00	0.024	0.032
20	FR1 n7	50M	BPSK	135	67	Left Cheek	0mm	Ant 0	Receiver on	507000	2535	22.71	24.00	1.346	0.03	0.072	0.097
	FR1 n7	50M	BPSK	135	67	Left Tilted	0mm	Ant 0	Receiver on	507000	2535	22.71	24.00	1.346	-0.14	0.039	0.052
	FR1 n8	20M	BPSK	50	25	Right Cheek	0mm	Ant 0	Receiver on	179500	897.5	23.22	24.00	1.197	0.16	0.097	0.116
	FR1 n8	20M	BPSK	50	25	Right Tilted	0mm	Ant 0	Receiver on	179500	897.5	23.22	24.00	1.197	-0.12	0.058	0.069
21	FR1 n8	20M	BPSK	50	25	Left Cheek	0mm	Ant 0	Receiver on	179500	897.5	23.22	24.00	1.197	0.06	0.105	0.126
	FR1 n8	20M	BPSK	50	25	Left Tilted	0mm	Ant 0	Receiver on	179500	897.5	23.22	24.00	1.197	0.11	0.058	0.069
	FR1 n20	20M	BPSK	50	25	Right Cheek	0mm	Ant 0	Receiver on	169400	847	23.22	24.00	1.197	0.15	0.121	0.145
	FR1 n20	20M	BPSK	50	25	Right Tilted	0mm	Ant 0	Receiver on	169400	847	23.22	24.00	1.197	-0.01	0.073	0.087
22	FR1 n20	20M	BPSK	50	25	Left Cheek	0mm	Ant 0	Receiver on	169400	847	23.22	24.00	1.197	0.03	0.123	0.147
	FR1 n20	20M	BPSK	50	25	Left Tilted	0mm	Ant 0	Receiver on	169400	847	23.22	24.00	1.197	-0.06	0.064	0.077
	FR1 n28	30M	BPSK	80	40	Right Cheek	0mm	Ant 0	Receiver on	145100	725.5	23.25	24.00	1.189	-0.02	0.102	0.121
	FR1 n28	30M	BPSK	80	40	Right Tilted	0mm	Ant 0	Receiver on	145100	725.5	23.25	24.00	1.189	-0.12	0.062	0.074
23	FR1 n28	30M	BPSK	80	40	Left Cheek	0mm	Ant 0	Receiver on	145100	725.5	23.25	24.00	1.189	0.08	0.115	0.137
	FR1 n28	30M	BPSK	80	40	Left Tilted	0mm	Ant 0	Receiver on	145100	725.5	23.25	24.00	1.189	-0.16	0.062	0.074
	FR1 n40	100M	BPSK	135	67	Right Cheek	0mm	Ant 0	Receiver on	470000	2350	23.21	24.00	1.199	0.18	0.060	0.072
	FR1 n40	100M	BPSK	135	67	Right Tilted	0mm	Ant 0	Receiver on	470000	2350	23.21	24.00	1.199	-0.08	0.038	0.046
24	FR1 n40	100M	BPSK	135	67	Left Cheek	0mm	Ant 0	Receiver on	470000	2350	23.21	24.00	1.199	0.01	0.096	0.115
	FR1 n40	100M	BPSK	135	67	Left Tilted	0mm	Ant 0	Receiver on	470000	2350	23.21	24.00	1.199	-0.18	0.063	0.076
	FR1 n41_HPUE	100M	BPSK	135	67	Right Cheek	0mm	Ant 4	Receiver on	518598	2592.99	24.93	25.50	1.14	0.04	0.308	0.351
	FR1 n41_HPUE	100M	BPSK	135	67	Right Tilted	0mm	Ant 4	Receiver on	518598	2592.99	24.93	25.50	1.14	0.05	0.261	0.298
25	FR1 n41_HPUE	100M	BPSK	135	67	Left Cheek	0mm	Ant 4	Receiver on	518598	2592.99	24.93	25.50	1.14	-0.11	0.923	1.052
	FR1 n41_HPUE	100M	BPSK	135	67	Left Cheek	0mm	Ant 4	Receiver on	509202	2546.01	24.92	25.50	1.143	0.16	0.811	0.927
	FR1 n41_HPUE	100M	BPSK	135	67	Left Cheek	0mm	Ant 4	Receiver on	528000	2640	24.87	25.50	1.156	0.17	0.807	0.933
	FR1 n41_HPUE	100M	BPSK	135	67	Left Tilted	0mm	Ant 4	Receiver on	518598	2592.99	24.93	25.50	1.14	0.08	0.649	0.740
	FR1 n77_HPUE	100M	BPSK	135	67	Right Cheek	0mm	Ant 4	Receiver on	650000	3750	25.31	26.00	1.172	-0.09	0.175	0.205
	FR1 n77_HPUE	100M	BPSK	135	67	Right Tilted	0mm	Ant 4	Receiver on	650000	3750	25.31	26.00	1.172	0.05	0.179	0.210
26	FR1 n77_HPUE	100M	BPSK	135	67	Left Cheek	0mm	Ant 4	Receiver on	650000	3750	25.31	26.00	1.172	-0.08	0.582	0.682
	FR1 n77_HPUE	100M	BPSK	135	67	Left Tilted	0mm	Ant 4	Receiver on	650000	3750	25.31	26.00	1.172	0.15	0.384	0.450
	FR1 n79_HPUE	100M	BPSK	135	67	Right Cheek	0mm	Ant 4	Receiver on	730000	4950	26.57	27.00	1.104	-0.11	0.116	0.128
	FR1 n79_HPUE	100M	BPSK	135	67	Right Tilted	0mm	Ant 4	Receiver on	730000	4950	26.57	27.00	1.104	0.15	0.118	0.130
27	FR1 n79_HPUE	100M	BPSK	135	67	Left Cheek	0mm	Ant 4	Receiver on	730000	4950	26.57	27.00	1.104	0.01	0.446	0.492
	FR1 n79_HPUE	100M	BPSK	135	67	Left Tilted	0mm	Ant 4	Receiver on	730000	4950	26.57	27.00	1.104	-0.17	0.235	0.259

【WLAN 2.4GHz_5GHz SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	WLAN2.4GHz	802.11b 1Mbps	Right Cheek	0mm	Ant 6	Receiver on	7	2442	18.27	20.00	1.489	99.3	1.007	-0.16	0.245	0.367
	WLAN2.4GHz	802.11b 1Mbps	Right Tilted	0mm	Ant 6	Receiver on	7	2442	18.27	20.00	1.489	99.3	1.007	-0.13	0.192	0.288
28	WLAN2.4GHz	802.11b 1Mbps	Left Cheek	0mm	Ant 6	Receiver on	7	2442	18.27	20.00	1.489	99.3	1.007	-0.15	0.554	0.831
	WLAN2.4GHz	802.11b 1Mbps	Left Tilted	0mm	Ant 6	Receiver on	7	2442	18.27	20.00	1.489	99.3	1.007	0.05	0.378	0.567
	WLAN5GHz	802.11n-HT20 MCS0	Right Cheek	0mm	Ant 6	Receiver on	64	5320	14.13	15.50	1.371	95.13	1.051	-0.00	0.048	0.069
	WLAN5GHz	802.11n-HT20 MCS0	Right Tilted	0mm	Ant 6	Receiver on	64	5320	14.13	15.50	1.371	95.13	1.051	-0.10	0.053	0.076
	WLAN5GHz	802.11n-HT20 MCS0	Left Cheek	0mm	Ant 6	Receiver on	64	5320	14.13	15.50	1.371	95.13	1.051	0.18	0.073	0.105
29	WLAN5GHz	802.11n-HT20 MCS0	Left Tilted	0mm	Ant 6	Receiver on	64	5320	14.13	15.50	1.371	95.13	1.051	0.08	0.083	0.120
	WLAN5GHz	802.11n-HT20 MCS0	Right Cheek	0mm	Ant 6	Receiver on	140	5700	14.85	16.50	1.462	95.13	1.051	-0.11	0.038	0.058
	WLAN5GHz	802.11n-HT20 MCS0	Right Tilted	0mm	Ant 6	Receiver on	140	5700	14.85	16.50	1.462	95.13	1.051	-0.03	0.053	0.081
	WLAN5GHz	802.11n-HT20 MCS0	Left Cheek	0mm	Ant 6	Receiver on	140	5700	14.85	16.50	1.462	95.13	1.051	-0.19	0.051	0.078
	WLAN5GHz	802.11n-HT20 MCS0	Left Tilted	0mm	Ant 6	Receiver on	140	5700	14.85	16.50	1.462	95.13	1.051	0.01	0.064	0.098
	WLAN5GHz	802.11n-HT40 MCS0	Right Cheek	0mm	Ant 6	Receiver on	151	5755	8.65	10	1.365	90.72	1.102	0.02	0.018	0.027
	WLAN5GHz	802.11n-HT40 MCS0	Right Tilted	0mm	Ant 6	Receiver on	151	5755	8.65	10	1.365	90.72	1.102	0.03	0.024	0.036
	WLAN5GHz	802.11n-HT40 MCS0	Left Cheek	0mm	Ant 6	Receiver on	151	5755	8.65	10	1.365	90.72	1.102	0.03	0.025	0.038
	WLAN5GHz	802.11n-HT40 MCS0	Left Tilted	0mm	Ant 6	Receiver on	151	5755	8.65	10	1.365	90.72	1.102	0.02	0.031	0.047

【Bluetooth SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	Bluetooth	1Mbps	Right Cheek	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.04	0.001	0.001
	Bluetooth	1Mbps	Right Tilted	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.14	0.001	0.001
30	Bluetooth	1Mbps	Left Cheek	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.04	0.030	0.045
	Bluetooth	1Mbps	Left Tilted	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.06	0.001	0.001

14.2 Body SAR

【GSM SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Headset	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
31	GSM900	GPRS 2 Tx slots	Front	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	-0.08	0.297	0.381
	GSM900	GPRS 2 Tx slots	Back	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	-0.05	0.593	0.760
	GSM900	GPRS 2 Tx slots	Back	5mm	Ant 0	Receiver off	37	897.4	Headset	30.42	31.50	1.282	0.04	0.555	0.712
	GSM900	GPRS 2 Tx slots	Left Side	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	0.05	0.088	0.113
	GSM900	GPRS 2 Tx slots	Right Side	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	-0.07	0.149	0.191
	GSM900	GPRS 2 Tx slots	Top Side	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	-0.15	0.001	0.001
	GSM900	GPRS 2 Tx slots	Bottom Side	5mm	Ant 0	Receiver off	37	897.4		30.42	31.50	1.282	0.14	0.332	0.426
	GSM1800	GPRS 3 Tx slots	Front	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	0.02	0.396	0.440
	GSM1800	GPRS 3 Tx slots	Back	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	-0.05	0.928	1.032
	GSM1800	GPRS 3 Tx slots	Back	5mm	Ant 0	Receiver off	512	1710.2		25.88	26.50	1.153	-0.18	0.863	0.995
	GSM1800	GPRS 3 Tx slots	Back	5mm	Ant 0	Receiver off	885	1784.8		25.81	26.50	1.172	0.17	0.672	0.788
	GSM1800	GPRS 3 Tx slots	Back	5mm	Ant 0	Receiver off	698	1747.4	Headset	26.04	26.50	1.112	-0.02	0.416	0.463
	GSM1800	GPRS 3 Tx slots	Left Side	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	-0.01	0.043	0.048
	GSM1800	GPRS 3 Tx slots	Right Side	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	-0.01	0.045	0.050
	GSM1800	GPRS 3 Tx slots	Top Side	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	0.05	0.001	0.001
	GSM1800	GPRS 3 Tx slots	Bottom Side	5mm	Ant 0	Receiver off	698	1747.4		26.04	26.50	1.112	-0.15	1.200	1.334
	GSM1800	GPRS 3 Tx slots	Bottom Side	5mm	Ant 0	Receiver off	512	1710.2		25.88	26.50	1.153	0.07	1.080	1.245
	GSM1800	GPRS 3 Tx slots	Bottom Side	5mm	Ant 0	Receiver off	885	1784.8		25.81	26.50	1.172	-0.12	0.926	1.085

【WCDMA SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Headset	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	WCDMA I	RMC 12.2Kbps	Front	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	0.15	0.399	0.560
	WCDMA I	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	-0.19	0.766	1.075
	WCDMA I	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	9612	1922.4		21.89	23.50	1.449	0.16	0.729	1.056
	WCDMA I	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	9888	1977.6		21.76	23.50	1.493	-0.11	0.697	1.041
	WCDMA I	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	9750	1950	Headset	22.03	23.50	1.403	-0.01	0.778	1.092
	WCDMA I	RMC 12.2Kbps	Left Side	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	0.07	0.143	0.201
	WCDMA I	RMC 12.2Kbps	Right Side	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	-0.02	0.042	0.059
	WCDMA I	RMC 12.2Kbps	Top Side	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	0.13	0.021	0.029
	WCDMA I	RMC 12.2Kbps	Bottom Side	5mm	Ant 0	Receiver off	9750	1950		22.03	23.50	1.403	-0.08	0.978	1.372
	WCDMA I	RMC 12.2Kbps	Bottom Side	5mm	Ant 0	Receiver off	9612	1922.4		21.89	23.50	1.449	0.05	1.010	1.463
	WCDMA I	RMC 12.2Kbps	Bottom Side	5mm	Ant 0	Receiver off	9888	1977.6		21.76	23.50	1.493	0.08	0.954	1.424
	WCDMA V	RMC 12.2Kbps	Front	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	-0.09	0.124	0.151
34	WCDMA V	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	-0.02	0.255	0.310
	WCDMA V	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	4182	836.4	Headset	22.15	23.00	1.216	-0.06	0.145	0.176
	WCDMA V	RMC 12.2Kbps	Left Side	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	-0.03	0.067	0.081
	WCDMA V	RMC 12.2Kbps	Right Side	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	0.08	0.099	0.120
	WCDMA V	RMC 12.2Kbps	Top Side	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	0.03	0.001	0.001
	WCDMA V	RMC 12.2Kbps	Bottom Side	5mm	Ant 0	Receiver off	4182	836.4		22.15	23.00	1.216	-0.15	0.150	0.182
35	WCDMA VIII	RMC 12.2Kbps	Front	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	-0.13	0.148	0.210
	WCDMA VIII	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	0.02	0.307	0.437
	WCDMA VIII	RMC 12.2Kbps	Back	5mm	Ant 0	Receiver off	2788	897.6	Headset	22.47	24.00	1.422	-0.03	0.222	0.316
	WCDMA VIII	RMC 12.2Kbps	Left Side	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	0.16	0.055	0.078
	WCDMA VIII	RMC 12.2Kbps	Right Side	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	-0.05	0.078	0.111
	WCDMA VIII	RMC 12.2Kbps	Top Side	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	0.18	0.001	0.001
	WCDMA VIII	RMC 12.2Kbps	Bottom Side	5mm	Ant 0	Receiver off	2788	897.6		22.47	24.00	1.422	0.04	0.221	0.314



[LTE SAR]

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Headset	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	LTE Band 1	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			0.13	0.401	0.532
	LTE Band 1	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			-0.18	0.656	0.871
	LTE Band 1	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	18300	1950	Headset	22.77	24.00	1.327			-0.04	0.640	0.849
	LTE Band 1	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			-0.14	0.168	0.223
	LTE Band 1	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			0.14	0.056	0.074
	LTE Band 1	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			-0.17	0.001	0.001
	LTE Band 1	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	18300	1950		22.77	24.00	1.327			0.13	0.972	1.290
	LTE Band 1	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	18100	1930		22.65	24.00	1.365			0.03	0.945	1.290
36	LTE Band 1	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	18500	1970		22.74	24.00	1.337			-0.09	0.972	1.300
	LTE Band 3	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			0.04	0.329	0.407
	LTE Band 3	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			-0.11	0.505	0.624
	LTE Band 3	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	19575	1747.5	Headset	22.08	23.00	1.236			0.12	0.485	0.599
	LTE Band 3	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			-0.13	0.080	0.099
	LTE Band 3	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			-0.01	0.039	0.048
	LTE Band 3	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			0.17	0.001	0.001
37	LTE Band 3	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	19575	1747.5		22.08	23.00	1.236			-0.13	0.699	0.864
	LTE Band 5	10M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			-0.07	0.246	0.296
38	LTE Band 5	10M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			-0.12	0.573	0.689
	LTE Band 5	10M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	20525	836.5	Headset	23.20	24.00	1.202			-0.18	0.383	0.460
	LTE Band 5	10M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			-0.17	0.045	0.054
	LTE Band 5	10M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			0.03	0.066	0.079
	LTE Band 5	10M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			-0.10	0.001	0.001
	LTE Band 5	10M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	20525	836.5		23.20	24.00	1.202			0.12	0.300	0.361
	LTE Band 7	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			-0.15	0.520	0.648
	LTE Band 7	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			-0.10	0.505	0.630
	LTE Band 7	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	21100	2535	Headset	22.54	23.50	1.247			0.08	0.465	0.580
	LTE Band 7	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			-0.07	0.265	0.330
	LTE Band 7	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			0.12	0.060	0.075
	LTE Band 7	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			0.12	0.025	0.031
39	LTE Band 7	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	21100	2535		22.54	23.50	1.247			-0.04	0.537	0.670
	LTE Band 8	10M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			0.04	0.301	0.387
40	LTE Band 8	10M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			0.12	0.712	0.915
	LTE Band 8	10M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	21625	897.5	Headset	22.91	24.00	1.285			0.14	0.687	0.883
	LTE Band 8	10M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			0.04	0.055	0.071
	LTE Band 8	10M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			-0.04	0.115	0.148
	LTE Band 8	10M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			-0.15	0.002	0.003
	LTE Band 8	10M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	21625	897.5		22.91	24.00	1.285			0.06	0.376	0.483
	LTE Band 20	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			-0.13	0.166	0.199
41	LTE Band 20	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			-0.07	0.349	0.418
	LTE Band 20	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	24300	847	Headset	23.21	24.00	1.199			-0.02	0.272	0.326
	LTE Band 20	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			-0.04	0.276	0.331
	LTE Band 20	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			-0.07	0.134	0.161
	LTE Band 20	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			-0.05	0.001	0.001
	LTE Band 20	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	24300	847		23.21	24.00	1.199			0.17	0.172	0.206
	LTE Band 26	15M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			0.10	0.330	0.433
	LTE Band 26	15M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			0.19	0.812	1.065
42	LTE Band 26	15M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	26765	821.5		22.54	24.00	1.4			0.19	0.812	1.137
	LTE Band 26	15M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	26965	841.5		22.51	24.00	1.409			0.10	0.786	1.107
	LTE Band 26	15M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	26865	831.5	Headset	22.82	24.00	1.312			-0.19	0.731	0.959
	LTE Band 26	15M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			0.19	0.110	0.144
	LTE Band 26	15M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			-0.14	0.184	0.241
	LTE Band 26	15M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			0.02	0.002	0.003
	LTE Band 26	15M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	26865	831.5		22.82	24.00	1.312			-0.05	0.330	0.433
	LTE Band 28	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			-0.03	0.123	0.150
43	LTE Band 28	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			-0.05	0.228	0.278
	LTE Band 28	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	27460	728	Headset	23.14	24.00	1.219			-0.06	0.151	0.184
	LTE Band 28	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			0.08	0.001	0.001
	LTE Band 28	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			-0.12	0.081	0.099
	LTE Band 28	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			0.01	0.001	0.001
	LTE Band 28	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	27460	728		23.14	24.00	1.219			-0.10	0.134	0.163
	LTE Band 40	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	0.05	0.476	0.601
	LTE Band 40	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	0.07	0.492	0.622
	LTE Band 40	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	39150	2350	Headset	23.01	24.00	1.256	62.9	1.006	-0.17	0.487	0.615
	LTE Band 40	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	0.01	0.146	0.184
	LTE Band 40	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	-0.08	0.092	0.116
	LTE Band 40	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	0.14	0.001	0.001
	LTE Band 40	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	39150	2350		23.01	24.00	1.256	62.9	1.006	-0.06	0.833	1.053



CE SAR Test Report

Report No. : EA612311

44	LTE Band 40	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	38750	2310		22.95	24.00	1.274	62.9	1.006	-0.10	0.960	1.230
	LTE Band 40	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	39550	2390		22.72	24.00	1.343	62.9	1.006	0.02	0.732	0.989
	LTE Band 41	20M	QPSK	1	0	Front	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	0.07	0.193	0.244
45	LTE Band 41	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	-0.05	0.224	0.283
	LTE Band 41	20M	QPSK	1	0	Back	5mm	Ant 0	Receiver off	40620	2593	Headset	23.01	24.00	1.256	62.9	1.006	-0.11	0.220	0.278
	LTE Band 41	20M	QPSK	1	0	Left Side	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	0.16	0.107	0.135
	LTE Band 41	20M	QPSK	1	0	Right Side	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	-0.04	0.026	0.033
	LTE Band 41	20M	QPSK	1	0	Top Side	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	-0.16	0.001	0.001
	LTE Band 41	20M	QPSK	1	0	Bottom Side	5mm	Ant 0	Receiver off	40620	2593		23.01	24.00	1.256	62.9	1.006	-0.16	0.185	0.234
	LTE Band 42	20M	QPSK	1	0	Front	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	0.05	0.091	0.118
46	LTE Band 42	20M	QPSK	1	0	Back	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	-0.04	0.234	0.305
	LTE Band 42	20M	QPSK	1	0	Back	5mm	Ant 4	Receiver off	42590	3500	Headset	22.88	24.00	1.294	62.9	1.006	0.14	0.172	0.224
	LTE Band 42	20M	QPSK	1	0	Left Side	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	-0.11	0.010	0.013
	LTE Band 42	20M	QPSK	1	0	Right Side	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	-0.09	0.179	0.233
	LTE Band 42	20M	QPSK	1	0	Top Side	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	-0.13	0.061	0.079
	LTE Band 42	20M	QPSK	1	0	Bottom Side	5mm	Ant 4	Receiver off	42590	3500		22.88	24.00	1.294	62.9	1.006	0.01	0.001	0.001

[FR1 SAR]

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Headset	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	FR1 n1	50M	BPSK	135	67	Front	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	0.17	0.444	0.554
	FR1 n1	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	0.18	0.802	1.000
	FR1 n1	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	389000	1945		22.36	23.50	1.3	-0.14	0.810	1.053
	FR1 n1	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	391000	1955		22.43	23.50	1.279	-0.05	0.795	1.017
	FR1 n1	50M	BPSK	135	67	Left Side	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	0.08	0.152	0.190
	FR1 n1	50M	BPSK	135	67	Right Side	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	-0.03	0.069	0.086
	FR1 n1	50M	BPSK	135	67	Top Side	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	0.18	0.021	0.026
	FR1 n1	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	390000	1950		22.54	23.50	1.247	0.01	1.180	1.471
47	FR1 n1	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	389000	1945		22.36	23.50	1.3	0.18	1.140	1.482
	FR1 n1	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	391000	1955		22.43	23.50	1.279	-0.09	1.120	1.432
	FR1 n1	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	390000	1950	Headset	22.54	23.50	1.247	0.14	0.750	0.935
	FR1 n3	50M	BPSK	135	67	Front	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	0.16	0.346	0.451
	FR1 n3	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	-0.05	0.817	1.065
	FR1 n3	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	347000	1735		22.31	23.50	1.315	-0.01	0.820	1.078
	FR1 n3	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	352000	1760		22.22	23.50	1.343	-0.18	0.837	1.124
	FR1 n3	50M	BPSK	135	67	Left Side	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	-0.08	0.037	0.048
	FR1 n3	50M	BPSK	135	67	Right Side	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	0.12	0.049	0.064
	FR1 n3	50M	BPSK	135	67	Top Side	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	0.13	0.001	0.001
	FR1 n3	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	349500	1747.5		22.35	23.50	1.303	0.05	1.100	1.433
	FR1 n3	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	347000	1735		22.31	23.50	1.315	-0.17	1.030	1.354
48	FR1 n3	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	352000	1760		22.22	23.50	1.343	-0.01	1.070	1.437
	FR1 n3	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	349500	1747.5	Headset	22.35	23.50	1.303	-0.12	0.438	0.571
	FR1 n5	20M	BPSK	50	25	Front	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	-0.01	0.182	0.216
49	FR1 n5	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	0.09	0.401	0.477
	FR1 n5	20M	BPSK	50	25	Left Side	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	0.17	0.095	0.113
	FR1 n5	20M	BPSK	50	25	Right Side	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	0.06	0.137	0.163
	FR1 n5	20M	BPSK	50	25	Top Side	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	0.10	0.001	0.001
	FR1 n5	20M	BPSK	50	25	Bottom Side	5mm	Ant 0	Receiver off	167300	836.5		23.25	24.00	1.189	0.15	0.229	0.272
	FR1 n5	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	167300	836.5	Headset	23.25	24.00	1.189	-0.08	0.305	0.363
	FR1 n7	50M	BPSK	135	67	Front	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	0.07	0.471	0.630
50	FR1 n7	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	0.02	0.532	0.711
	FR1 n7	50M	BPSK	135	67	Left Side	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	0.06	0.249	0.333
	FR1 n7	50M	BPSK	135	67	Right Side	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	0.09	0.058	0.078
	FR1 n7	50M	BPSK	135	67	Top Side	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	-0.11	0.024	0.032
	FR1 n7	50M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	507000	2535		21.24	22.50	1.337	0.17	0.527	0.705
	FR1 n7	50M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	507000	2535	Headset	21.24	22.50	1.337	-0.15	0.476	0.636
	FR1 n8	20M	BPSK	50	25	Front	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	0.15	0.245	0.293
51	FR1 n8	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	0.09	0.476	0.570
	FR1 n8	20M	BPSK	50	25	Left Side	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	0.02	0.094	0.113
	FR1 n8	20M	BPSK	50	25	Right Side	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	0.06	0.143	0.171
	FR1 n8	20M	BPSK	50	25	Top Side	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	0.05	0.001	0.001
	FR1 n8	20M	BPSK	50	25	Bottom Side	5mm	Ant 0	Receiver off	179500	897.5		23.22	24.00	1.197	-0.05	0.316	0.378
	FR1 n8	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	179500	897.5	Headset	23.22	24.00	1.197	-0.04	0.433	0.518
	FR1 n20	20M	BPSK	50	25	Front	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	-0.17	0.188	0.225
52	FR1 n20	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	0.01	0.401	0.480
	FR1 n20	20M	BPSK	50	25	Left Side	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	0.12	0.097	0.116
	FR1 n20	20M	BPSK	50	25	Right Side	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	0.11	0.142	0.170
	FR1 n20	20M	BPSK	50	25	Top Side	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	-0.16	0.001	0.001
	FR1 n20	20M	BPSK	50	25	Bottom Side	5mm	Ant 0	Receiver off	169400	847		23.22	24.00	1.197	-0.12	0.253	0.303
	FR1 n20	20M	BPSK	50	25	Back	5mm	Ant 0	Receiver off	169400	847	Headset	23.22	24.00	1.197	-0.08	0.288	0.345



CE SAR Test Report

Report No. : EA612311

	FR1 n28	30M	BPSK	80	40	Front	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	0.05	0.146	0.174
53	FR1 n28	30M	BPSK	80	40	Back	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	0.19	0.271	0.322
	FR1 n28	30M	BPSK	80	40	Left Side	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	0.07	0.129	0.153
	FR1 n28	30M	BPSK	80	40	Right Side	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	-0.12	0.198	0.235
	FR1 n28	30M	BPSK	80	40	Top Side	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	-0.17	0.001	0.001
	FR1 n28	30M	BPSK	80	40	Bottom Side	5mm	Ant 0	Receiver off	145100	725.5		23.25	24.00	1.189	0.01	0.171	0.203
	FR1 n28	30M	BPSK	80	40	Back	5mm	Ant 0	Receiver off	145100	725.5	Headset	23.25	24.00	1.189	-0.13	0.204	0.243
	FR1 n40	100M	BPSK	135	67	Front	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	-0.08	0.614	0.738
	FR1 n40	100M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	-0.03	0.791	0.951
	FR1 n40	100M	BPSK	135	67	Left Side	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	0.06	0.150	0.180
	FR1 n40	100M	BPSK	135	67	Right Side	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	0.13	0.119	0.143
	FR1 n40	100M	BPSK	135	67	Top Side	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	0.18	0.002	0.002
54	FR1 n40	100M	BPSK	135	67	Bottom Side	5mm	Ant 0	Receiver off	470000	2350		22.20	23.00	1.202	0.06	1.210	1.454
	FR1 n40	100M	BPSK	135	67	Back	5mm	Ant 0	Receiver off	470000	2350	Headset	22.20	23.00	1.202	0.19	0.771	0.927
	FR1 n41_HPUE	100M	BPSK	135	67	Front	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	0.03	0.330	0.374
	FR1 n41_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	0.10	0.953	1.079
	FR1 n41_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	509202	2546.01		23.94	24.50	1.138	-0.19	0.972	1.106
55	FR1 n41_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	528000	2640		23.84	24.50	1.164	-0.11	1.100	1.280
	FR1 n41_HPUE	100M	BPSK	135	67	Left Side	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	-0.15	0.020	0.023
	FR1 n41_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	-0.10	0.412	0.466
	FR1 n41_HPUE	100M	BPSK	135	67	Top Side	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	-0.02	0.227	0.257
	FR1 n41_HPUE	100M	BPSK	135	67	Bottom Side	5mm	Ant 4	Receiver off	518598	2592.99		23.96	24.50	1.132	0.06	0.001	0.001
	FR1 n41_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	518598	2592.99	Headset	23.96	24.50	1.132	0.01	0.953	1.079
	FR1 n77_HPUE	100M	BPSK	135	67	Front	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	-0.07	0.384	0.450
56	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	-0.04	1.060	1.242
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	623334	3350.01		25.27	26.00	1.183	-0.18	0.958	1.133
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	633332	3499.98		25.24	26.00	1.191	0.15	0.631	0.752
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	640000	3600		25.03	26.00	1.25	0.04	0.762	0.953
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	660000	3900		24.62	26.00	1.374	-0.12	0.833	1.145
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	666666	3999.99		25.24	26.00	1.191	0.13	0.917	1.092
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	676666	4149.99		24.48	26.00	1.419	0.03	0.829	1.176
	FR1 n77_HPUE	100M	BPSK	135	67	Left Side	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	-0.05	0.002	0.002
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	0.18	0.719	0.843
	FR1 n77_HPUE	100M	BPSK	135	67	Top Side	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	-0.01	0.308	0.361
	FR1 n77_HPUE	100M	BPSK	135	67	Bottom Side	5mm	Ant 4	Receiver off	650000	3750		25.31	26.00	1.172	-0.03	0.002	0.002
	FR1 n77_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	650000	3750	Headset	25.31	26.00	1.172	-0.09	0.903	1.058
	FR1 n79_HPUE	100M	BPSK	135	67	Front	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	0.06	0.221	0.241
	FR1 n79_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	-0.05	0.337	0.367
	FR1 n79_HPUE	100M	BPSK	135	67	Left Side	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	0.18	0.002	0.002
57	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	-0.05	0.962	1.048
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	696668	4450.02		21.93	23.50	1.435	0.06	0.721	1.035
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	705000	4575		21.92	23.50	1.439	-0.04	0.684	0.984
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	713332	4699.98		22.48	23.50	1.265	0.03	0.706	0.893
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	5mm	Ant 4	Receiver off	721666	4824.99		22.37	23.50	1.297	-0.03	0.689	0.894
	FR1 n79_HPUE	100M	BPSK	135	67	Top Side	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	0.11	0.219	0.238
	FR1 n79_HPUE	100M	BPSK	135	67	Bottom Side	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	-0.07	0.002	0.002
	FR1 n79_HPUE	100M	BPSK	135	67	Back	5mm	Ant 4	Receiver off	730000	4950		23.13	23.50	1.089	0.12	0.323	0.352

[WLAN 2.4GHz_5GHz SAR]

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Headset	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
58	WLAN2.4GHz	802.11b 1Mbps	Front	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	0.09	0.216	0.324
	WLAN2.4GHz	802.11b 1Mbps	Back	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	-0.15	0.467	0.700
	WLAN2.4GHz	802.11b 1Mbps	Left Side	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	0.11	0.001	0.001
	WLAN2.4GHz	802.11b 1Mbps	Right Side	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	0.04	0.166	0.249
	WLAN2.4GHz	802.11b 1Mbps	Top Side	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	-0.09	0.153	0.229
	WLAN2.4GHz	802.11b 1Mbps	Bottom Side	5mm	Ant 6	Receiver off	7	2442		18.27	20.00	1.489	99.3	1.007	0.15	0.001	0.001
	WLAN2.4GHz	802.11b 1Mbps	Back	5mm	Ant 6	Receiver off	7	2442	Headset	18.27	20.00	1.489	99.3	1.007	0.07	0.447	0.670
	WLAN5GHz	802.11n-HT20 MCS0	Front	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	-0.19	0.052	0.075
	WLAN5GHz	802.11n-HT20 MCS0	Back	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	-0.17	0.135	0.195
	WLAN5GHz	802.11n-HT20 MCS0	Left Side	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	0.09	0.001	0.001
	WLAN5GHz	802.11n-HT20 MCS0	Right Side	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	-0.04	0.091	0.131
	WLAN5GHz	802.11n-HT20 MCS0	Top Side	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	0.06	0.183	0.264
	WLAN5GHz	802.11n-HT20 MCS0	Bottom Side	5mm	Ant 6	Receiver off	64	5320		14.13	15.50	1.371	95.13	1.051	-0.07	0.001	0.001
59	WLAN5GHz	802.11n-HT20 MCS0	Back	5mm	Ant 6	Receiver off	64	5320	Headset	14.13	15.50	1.371	95.13	1.051	-0.14	0.120	0.173
	WLAN5GHz	802.11n-HT20 MCS0	Front	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	-0.06	0.033	0.051
	WLAN5GHz	802.11n-HT20 MCS0	Back	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	0.17	0.145	0.223
	WLAN5GHz	802.11n-HT20 MCS0	Left Side	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	0.03	0.017	0.026
	WLAN5GHz	802.11n-HT20 MCS0	Right Side	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	0.05	0.065	0.100
	WLAN5GHz	802.11n-HT20 MCS0	Top Side	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	-0.04	0.161	0.247
	WLAN5GHz	802.11n-HT20 MCS0	Bottom Side	5mm	Ant 6	Receiver off	140	5700		14.85	16.50	1.462	95.13	1.051	-0.15	0.001	0.002
	WLAN5GHz	802.11n-HT20 MCS0	Back	5mm	Ant 6	Receiver off	140	5700	Headset	14.85	16.50	1.462	95.13	1.051	0.12	0.143	0.220
	WLAN5GHz	802.11n-HT40 MCS0	Front	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.03	0.015	0.023
	WLAN5GHz	802.11n-HT40 MCS0	Back	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.02	0.054	0.081
	WLAN5GHz	802.11n-HT40 MCS0	Left Side	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.01	0.007	0.011
	WLAN5GHz	802.11n-HT40 MCS0	Right Side	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.05	0.028	0.042
	WLAN5GHz	802.11n-HT40 MCS0	Top Side	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.06	0.071	0.107
	WLAN5GHz	802.11n-HT40 MCS0	Bottom Side	5mm	Ant 6	Receiver off	151	5755		8.65	10	1.365	90.72	1.102	0.08	0.001	0.002
	WLAN5GHz	802.11n-HT40 MCS0	Back	5mm	Ant 6	Receiver off	151	5755	Headset	8.65	10	1.365	90.72	1.102	0.02	0.035	0.053

[Bluetooth SAR]

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
60	Bluetooth	1Mbps	Front	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.05	0.001	0.001
	Bluetooth	1Mbps	Back	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.18	0.041	0.061
	Bluetooth	1Mbps	Left Side	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.18	0.001	0.001
	Bluetooth	1Mbps	Right Side	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.04	0.001	0.001
	Bluetooth	1Mbps	Top Side	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.09	0.001	0.001
	Bluetooth	1Mbps	Bottom Side	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.04	0.001	0.001
	Bluetooth	1Mbps	Back	5mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.16	0.040	0.059

14.3 Limbs SAR

【GSM SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	GSM900	GPRS 2 Tx slots	Front	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	0.00	0.822	1.054
	GSM900	GPRS 2 Tx slots	Back	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	0.15	1.320	1.692
	GSM900	GPRS 2 Tx slots	Left Side	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	-0.14	0.086	0.110
	GSM900	GPRS 2 Tx slots	Right Side	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	0.04	0.337	0.432
	GSM900	GPRS 2 Tx slots	Top Side	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	-0.17	0.016	0.021
61	GSM900	GPRS 2 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	37	897.4	30.42	31.50	1.282	0.06	1.760	2.256
	GSM900	GPRS 2 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	975	880.2	30.23	31.50	1.34	-0.03	1.650	2.211
	GSM900	GPRS 2 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	124	914.8	30.01	31.50	1.409	0.01	1.210	1.705
	GSM1800	GPRS 3 Tx slots	Front	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	0.18	1.030	1.145
	GSM1800	GPRS 3 Tx slots	Back	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	-0.04	1.910	2.124
	GSM1800	GPRS 3 Tx slots	Back	0mm	Ant 0	Receiver off	512	1710.2	25.88	26.50	1.153	0.18	2.160	2.490
	GSM1800	GPRS 3 Tx slots	Back	0mm	Ant 0	Receiver off	885	1784.8	25.81	26.50	1.172	-0.02	1.460	1.711
	GSM1800	GPRS 3 Tx slots	Left Side	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	-0.17	0.222	0.247
	GSM1800	GPRS 3 Tx slots	Right Side	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	-0.18	0.094	0.105
	GSM1800	GPRS 3 Tx slots	Top Side	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	-0.07	0.001	0.001
62	GSM1800	GPRS 3 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	698	1747.4	26.04	26.50	1.112	0.05	2.300	2.558
	GSM1800	GPRS 3 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	512	1710.2	25.88	26.50	1.153	0.13	2.160	2.490
	GSM1800	GPRS 3 Tx slots	Bottom Side	0mm	Ant 0	Receiver off	885	1784.8	25.81	26.50	1.172	0.08	1.440	1.688

【WCDMA SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	WCDMA I	RMC 12.2Kbps	Front	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	0.07	1.040	1.459
	WCDMA I	RMC 12.2Kbps	Back	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	0.02	1.960	2.750
	WCDMA I	RMC 12.2Kbps	Back	0mm	Ant 0	Receiver off	9612	1922.4	21.89	23.50	1.449	0.12	2.050	2.970
	WCDMA I	RMC 12.2Kbps	Back	0mm	Ant 0	Receiver off	9888	1977.6	21.76	23.50	1.493	0.13	1.890	2.822
	WCDMA I	RMC 12.2Kbps	Left Side	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	0.03	0.501	0.703
	WCDMA I	RMC 12.2Kbps	Right Side	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	0.12	0.086	0.121
	WCDMA I	RMC 12.2Kbps	Top Side	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	-0.13	0.017	0.024
63	WCDMA I	RMC 12.2Kbps	Bottom Side	0mm	Ant 0	Receiver off	9750	1950	22.03	23.50	1.403	0.04	2.220	3.115
	WCDMA I	RMC 12.2Kbps	Bottom Side	0mm	Ant 0	Receiver off	9612	1922.4	21.89	23.50	1.449	-0.05	2.150	3.115
	WCDMA I	RMC 12.2Kbps	Bottom Side	0mm	Ant 0	Receiver off	9888	1977.6	21.76	23.50	1.493	0.07	2.050	3.061
	WCDMA V	RMC 12.2Kbps	Front	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	0.11	0.526	0.640
	WCDMA V	RMC 12.2Kbps	Back	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	0.06	0.760	0.924
	WCDMA V	RMC 12.2Kbps	Left Side	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	0.08	0.087	0.106
	WCDMA V	RMC 12.2Kbps	Right Side	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	0.04	0.268	0.326
	WCDMA V	RMC 12.2Kbps	Top Side	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	-0.04	0.019	0.023
64	WCDMA V	RMC 12.2Kbps	Bottom Side	0mm	Ant 0	Receiver off	4182	836.4	22.15	23.00	1.216	0.05	1.050	1.277
	WCDMA VIII	RMC 12.2Kbps	Front	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.17	0.438	0.623
	WCDMA VIII	RMC 12.2Kbps	Back	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.14	0.669	0.951
	WCDMA VIII	RMC 12.2Kbps	Left Side	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.12	0.068	0.097
	WCDMA VIII	RMC 12.2Kbps	Right Side	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.10	0.244	0.347
	WCDMA VIII	RMC 12.2Kbps	Top Side	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.12	0.001	0.001
65	WCDMA VIII	RMC 12.2Kbps	Bottom Side	0mm	Ant 0	Receiver off	2788	897.6	22.47	24.00	1.422	0.08	0.853	1.213



[LTE SAR]

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	LTE Band 1	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			-0.10	1.110	1.473
	LTE Band 1	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			0.01	1.750	2.322
	LTE Band 1	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	18100	1930	22.65	24.00	1.365			0.18	1.080	1.474
	LTE Band 1	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	18500	1970	22.74	24.00	1.337			0.14	1.780	2.380
	LTE Band 1	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			0.12	0.381	0.506
	LTE Band 1	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			-0.03	0.098	0.130
	LTE Band 1	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			0.18	0.032	0.042
66	LTE Band 1	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	18300	1950	22.77	24.00	1.327			0.01	2.010	2.667
	LTE Band 1	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	18100	1930	22.65	24.00	1.365			-0.02	1.260	1.720
	LTE Band 1	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	18500	1970	22.74	24.00	1.337			-0.06	1.320	1.765
	LTE Band 3	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			0.05	1.310	1.619
	LTE Band 3	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			-0.04	2.030	2.509
67	LTE Band 3	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	19300	1720	21.80	23.00	1.318			0.12	2.060	2.715
	LTE Band 3	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	19850	1775	21.74	23.00	1.337			-0.08	1.710	2.286
	LTE Band 3	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			0.08	0.420	0.519
	LTE Band 3	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			0.09	0.065	0.080
	LTE Band 3	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			0.16	0.022	0.027
	LTE Band 3	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	19575	1747.5	22.08	23.00	1.236			0.10	1.570	1.941
	LTE Band 5	10M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			-0.01	0.324	0.389
	LTE Band 5	10M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			-0.09	0.633	0.761
	LTE Band 5	10M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			-0.15	0.064	0.077
	LTE Band 5	10M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			-0.02	0.208	0.250
	LTE Band 5	10M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			-0.06	0.016	0.019
68	LTE Band 5	10M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	20525	836.5	23.20	24.00	1.202			0.09	1.030	1.238
	LTE Band 7	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			-0.05	1.270	1.584
	LTE Band 7	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			0.15	2.190	2.731
69	LTE Band 7	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	20850	2510	22.06	23.50	1.393			-0.09	2.260	3.148
	LTE Band 7	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	21350	2560	22.13	23.50	1.371			-0.15	1.480	2.029
	LTE Band 7	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			0.00	0.136	0.170
	LTE Band 7	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			-0.08	0.920	1.147
	LTE Band 7	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			-0.03	0.022	0.027
	LTE Band 7	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	21100	2535	22.54	23.50	1.247			0.17	1.400	1.746
	LTE Band 8	10M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			-0.18	0.192	0.247
	LTE Band 8	10M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			0.17	0.579	0.744
	LTE Band 8	10M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			-0.11	0.051	0.066
	LTE Band 8	10M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			-0.14	0.255	0.328
	LTE Band 8	10M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			0.13	0.001	0.001
70	LTE Band 8	10M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	21625	897.5	22.91	24.00	1.285			0.04	0.785	1.009
	LTE Band 20	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			0.02	0.387	0.464
	LTE Band 20	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			-0.11	0.553	0.663
	LTE Band 20	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			0.12	0.053	0.064
	LTE Band 20	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			-0.15	0.210	0.252
	LTE Band 20	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			-0.01	0.033	0.040
71	LTE Band 20	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	24300	847	23.21	24.00	1.199			0.02	0.888	1.065
	LTE Band 26	15M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			0.05	0.524	0.687
	LTE Band 26	15M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			-0.18	0.848	1.113
	LTE Band 26	15M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			0.06	0.073	0.096
	LTE Band 26	15M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			-0.04	0.246	0.323
	LTE Band 26	15M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			-0.09	0.036	0.047
72	LTE Band 26	15M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	26865	831.5	22.82	24.00	1.312			0.06	1.220	1.601
	LTE Band 28	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.17	0.561	0.684
	LTE Band 28	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.02	0.585	0.713
	LTE Band 28	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.03	0.078	0.095
	LTE Band 28	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.02	0.291	0.355
	LTE Band 28	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.01	0.024	0.029
73	LTE Band 28	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	27460	728	23.14	24.00	1.219			0.03	0.860	1.048
	LTE Band 40	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	0.08	1.130	1.428
74	LTE Band 40	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	0.06	2.090	2.641
	LTE Band 40	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	38750	2310	22.95	24.00	1.274	62.9	1.006	0.14	1.910	2.448
	LTE Band 40	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	39550	2390	22.72	24.00	1.343	62.9	1.006	0.15	1.850	2.499
	LTE Band 40	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	-0.01	0.477	0.603
	LTE Band 40	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	-0.19	0.168	0.212
	LTE Band 40	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	0.18	0.025	0.032
	LTE Band 40	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	39150	2350	23.01	24.00	1.256	62.9	1.006	0.07	1.360	1.718
	LTE Band 41	20M	QPSK	1	0	Front	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.10	0.758	0.958
75	LTE Band 41	20M	QPSK	1	0	Back	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	0.01	1.570	1.984
	LTE Band 41	20M	QPSK	1	0	Left Side	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	0.14	0.597	0.754



CE SAR Test Report

Report No. : EA612311

	LTE Band 41	20M	QPSK	1	0	Right Side	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.07	0.086	0.109
	LTE Band 41	20M	QPSK	1	0	Top Side	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.18	0.001	0.001
	LTE Band 41	20M	QPSK	1	0	Bottom Side	0mm	Ant 0	Receiver off	40620	2593	23.01	24.00	1.256	62.9	1.006	-0.07	0.890	1.125
	LTE Band 42	20M	QPSK	1	0	Front	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	0.06	0.176	0.229
76	LTE Band 42	20M	QPSK	1	0	Back	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.04	0.451	0.587
	LTE Band 42	20M	QPSK	1	0	Left Side	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	0.07	0.011	0.014
	LTE Band 42	20M	QPSK	1	0	Right Side	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	0.14	0.410	0.534
	LTE Band 42	20M	QPSK	1	0	Top Side	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	0.02	0.126	0.164
	LTE Band 42	20M	QPSK	1	0	Bottom Side	0mm	Ant 4	Receiver off	42590	3500	22.88	24.00	1.294	62.9	1.006	-0.08	0.001	0.001

【FR1 SAR】

Plot No	Band	BW	Modulation	RB Size	RB Offset	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	FR1 n1	50M	BPSK	135	67	Front	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	-0.09	1.100	1.372
	FR1 n1	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	0.13	2.190	2.731
	FR1 n1	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	389000	1945	22.36	23.50	1.3	-0.06	2.080	2.704
	FR1 n1	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	391000	1955	22.43	23.50	1.279	-0.04	2.070	2.648
	FR1 n1	50M	BPSK	135	67	Left Side	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	0.06	0.501	0.625
	FR1 n1	50M	BPSK	135	67	Right Side	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	0.08	0.100	0.125
	FR1 n1	50M	BPSK	135	67	Top Side	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	-0.15	0.029	0.036
	FR1 n1	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	390000	1950	22.54	23.50	1.247	-0.04	2.470	3.080
77	FR1 n1	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	389000	1945	22.36	23.50	1.3	-0.11	2.410	3.133
	FR1 n1	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	391000	1955	22.43	23.50	1.279	-0.14	2.410	3.082
	FR1 n3	50M	BPSK	135	67	Front	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	-0.13	1.060	1.381
	FR1 n3	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	0.09	2.370	3.088
	FR1 n3	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	347000	1735	22.31	23.50	1.315	0.18	2.300	3.024
78	FR1 n3	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	352000	1760	22.22	23.50	1.343	-0.11	2.310	3.102
	FR1 n3	50M	BPSK	135	67	Left Side	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	-0.19	0.228	0.297
	FR1 n3	50M	BPSK	135	67	Right Side	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	0.07	0.100	0.130
	FR1 n3	50M	BPSK	135	67	Top Side	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	0.06	0.001	0.001
	FR1 n3	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	349500	1747.5	22.35	23.50	1.303	0.17	2.250	2.932
	FR1 n3	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	347000	1735	22.31	23.50	1.315	0.14	2.230	2.932
	FR1 n3	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	352000	1760	22.22	23.50	1.343	-0.00	2.290	3.075
	FR1 n5	20M	BPSK	50	25	Front	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	-0.00	0.499	0.593
	FR1 n5	20M	BPSK	50	25	Back	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	-0.16	0.748	0.889
	FR1 n5	20M	BPSK	50	25	Left Side	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	0.02	0.075	0.089
	FR1 n5	20M	BPSK	50	25	Right Side	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	0.07	0.266	0.316
	FR1 n5	20M	BPSK	50	25	Top Side	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	-0.18	0.017	0.020
79	FR1 n5	20M	BPSK	50	25	Bottom Side	0mm	Ant 0	Receiver off	167300	836.5	23.25	24.00	1.189	0.08	1.170	1.391
	FR1 n7	50M	BPSK	135	67	Front	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	0.09	0.925	1.237
	FR1 n7	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	0.17	1.990	2.661
80	FR1 n7	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	505000	2525	21.14	22.50	1.368	0.06	2.090	2.859
	FR1 n7	50M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	509000	2545	21.13	22.50	1.371	-0.00	2.000	2.742
	FR1 n7	50M	BPSK	135	67	Left Side	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	-0.16	0.722	0.965
	FR1 n7	50M	BPSK	135	67	Right Side	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	-0.14	0.119	0.159
	FR1 n7	50M	BPSK	135	67	Top Side	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	0.18	0.024	0.032
	FR1 n7	50M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	507000	2535	21.24	22.50	1.337	-0.14	1.230	1.645
	FR1 n8	20M	BPSK	50	25	Front	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	-0.18	0.589	0.705
	FR1 n8	20M	BPSK	50	25	Back	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	-0.17	0.824	0.986
	FR1 n8	20M	BPSK	50	25	Left Side	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	-0.10	0.089	0.107
	FR1 n8	20M	BPSK	50	25	Right Side	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	0.08	0.359	0.430
	FR1 n8	20M	BPSK	50	25	Top Side	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	-0.06	0.017	0.020
81	FR1 n8	20M	BPSK	50	25	Bottom Side	0mm	Ant 0	Receiver off	179500	897.5	23.22	24.00	1.197	0.08	1.100	1.317
	FR1 n20	20M	BPSK	50	25	Front	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	0.12	0.503	0.602
	FR1 n20	20M	BPSK	50	25	Back	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	-0.19	0.749	0.897
	FR1 n20	20M	BPSK	50	25	Left Side	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	0.12	0.079	0.095
	FR1 n20	20M	BPSK	50	25	Right Side	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	0.03	0.269	0.322
	FR1 n20	20M	BPSK	50	25	Top Side	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	-0.07	0.022	0.026
82	FR1 n20	20M	BPSK	50	25	Bottom Side	0mm	Ant 0	Receiver off	169400	847	23.22	24.00	1.197	0.19	1.160	1.389
	FR1 n28	30M	BPSK	80	40	Front	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	0.09	0.375	0.446
	FR1 n28	30M	BPSK	80	40	Back	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	0.07	0.450	0.535
	FR1 n28	30M	BPSK	80	40	Left Side	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	0.10	0.053	0.063
	FR1 n28	30M	BPSK	80	40	Right Side	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	0.17	0.156	0.185
	FR1 n28	30M	BPSK	80	40	Top Side	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	-0.00	0.013	0.015
83	FR1 n28	30M	BPSK	80	40	Bottom Side	0mm	Ant 0	Receiver off	145100	725.5	23.25	24.00	1.189	0.07	0.742	0.882
	FR1 n40	100M	BPSK	135	67	Front	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	0.09	1.440	1.731
84	FR1 n40	100M	BPSK	135	67	Back	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	0.07	2.450	2.945
	FR1 n40	100M	BPSK	135	67	Left Side	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	0.07	0.344	0.413
	FR1 n40	100M	BPSK	135	67	Right Side	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	0.03	0.196	0.236
	FR1 n40	100M	BPSK	135	67	Top Side	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	-0.10	0.025	0.030



CE SAR Test Report

Report No. : EA612311

	FR1 n40	100M	BPSK	135	67	Bottom Side	0mm	Ant 0	Receiver off	470000	2350	22.20	23.00	1.202	-0.14	1.640	1.971
	FR1 n41_HPUE	100M	BPSK	135	67	Front	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	-0.09	1.340	1.517
	FR1 n41_HPUE	100M	BPSK	135	67	Back	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	0.07	2.220	2.513
	FR1 n41_HPUE	100M	BPSK	135	67	Back	0mm	Ant 4	Receiver off	509202	2546.01	23.94	24.50	1.138	0.12	2.560	2.913
85	FR1 n41_HPUE	100M	BPSK	135	67	Back	0mm	Ant 4	Receiver off	528000	2640	23.84	24.50	1.164	-0.02	2.670	3.108
	FR1 n41_HPUE	100M	BPSK	135	67	Left Side	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	0.09	0.079	0.089
	FR1 n41_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	0.12	1.550	1.755
	FR1 n41_HPUE	100M	BPSK	135	67	Top Side	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	0.05	0.658	0.745
	FR1 n41_HPUE	100M	BPSK	135	67	Bottom Side	0mm	Ant 4	Receiver off	518598	2592.99	23.96	24.50	1.132	0.01	0.024	0.027
	FR1 n77_HPUE	100M	BPSK	135	67	Front	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	-0.08	1.110	1.301
	FR1 n77_HPUE	100M	BPSK	135	67	Back	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	-0.12	1.660	1.946
	FR1 n77_HPUE	100M	BPSK	135	67	Left Side	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	0.08	0.037	0.043
86	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	-0.15	2.410	2.825
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	623334	3350.01	25.27	26.00	1.183	0.16	1.990	2.354
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	633332	3499.98	25.24	26.00	1.191	0.17	1.310	1.560
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	640000	3600	25.03	26.00	1.25	-0.10	1.810	1.975
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	660000	3900	24.62	26.00	1.374	-0.07	1.730	2.377
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	666666	3999.99	25.24	26.00	1.191	-0.01	1.910	2.275
	FR1 n77_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	676666	4149.99	24.48	26.00	1.419	-0.08	1.720	2.441
	FR1 n77_HPUE	100M	BPSK	135	67	Top Side	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	0.16	1.350	1.582
	FR1 n77_HPUE	100M	BPSK	135	67	Bottom Side	0mm	Ant 4	Receiver off	650000	3750	25.31	26.00	1.172	0.05	0.058	0.068
	FR1 n79_HPUE	100M	BPSK	135	67	Front	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	-0.01	0.302	0.329
	FR1 n79_HPUE	100M	BPSK	135	67	Back	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	-0.18	0.537	0.585
	FR1 n79_HPUE	100M	BPSK	135	67	Left Side	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	-0.17	0.001	0.001
87	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	-0.03	2.870	3.125
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	696668	4450.02	21.93	23.50	1.435	-0.15	0.454	0.651
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	705000	4575	21.92	23.50	1.439	0.05	0.536	0.771
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	713332	4699.98	22.48	23.50	1.265	-0.02	1.490	1.885
	FR1 n79_HPUE	100M	BPSK	135	67	Right Side	0mm	Ant 4	Receiver off	721666	4824.99	22.37	23.50	1.297	-0.09	2.180	2.827
	FR1 n79_HPUE	100M	BPSK	135	67	Top Side	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	0.11	0.270	0.294
	FR1 n79_HPUE	100M	BPSK	135	67	Bottom Side	0mm	Ant 4	Receiver off	730000	4950	23.13	23.50	1.089	0.10	0.001	0.001

【WLAN 2.4GHz_5GHz SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
	WLAN2.4GHz	802.11b 1Mbps	Front	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	0.19	0.861	1.291
88	WLAN2.4GHz	802.11b 1Mbps	Back	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	0.06	1.110	1.664
	WLAN2.4GHz	802.11b 1Mbps	Left Side	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	-0.08	0.002	0.003
	WLAN2.4GHz	802.11b 1Mbps	Right Side	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	-0.10	0.542	0.813
	WLAN2.4GHz	802.11b 1Mbps	Top Side	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	-0.02	0.776	1.164
	WLAN2.4GHz	802.11b 1Mbps	Bottom Side	0mm	Ant 6	Receiver off	7	2442	18.27	20.00	1.489	99.3	1.007	-0.08	0.002	0.003
	WLAN5GHz	802.11n-HT20 MCS0	Front	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	0.10	0.122	0.176
	WLAN5GHz	802.11n-HT20 MCS0	Back	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	0.19	0.202	0.291
	WLAN5GHz	802.11n-HT20 MCS0	Left Side	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	0.13	0.001	0.001
	WLAN5GHz	802.11n-HT20 MCS0	Right Side	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	0.11	0.187	0.269
	WLAN5GHz	802.11n-HT20 MCS0	Top Side	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	0.16	0.456	0.657
	WLAN5GHz	802.11n-HT20 MCS0	Bottom Side	0mm	Ant 6	Receiver off	64	5320	14.13	15.50	1.371	95.13	1.051	-0.05	0.001	0.001
	WLAN5GHz	802.11n-HT20 MCS0	Front	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	0.15	0.125	0.192
	WLAN5GHz	802.11n-HT20 MCS0	Back	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	0.01	0.282	0.433
	WLAN5GHz	802.11n-HT20 MCS0	Left Side	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	0.09	0.037	0.057
	WLAN5GHz	802.11n-HT20 MCS0	Right Side	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	0.11	0.199	0.306
89	WLAN5GHz	802.11n-HT20 MCS0	Top Side	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	0.04	0.475	0.730
	WLAN5GHz	802.11n-HT20 MCS0	Bottom Side	0mm	Ant 6	Receiver off	140	5700	14.85	16.50	1.462	95.13	1.051	-0.15	0.001	0.002
	WLAN5GHz	802.11n-HT40 MCS0	Front	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.05	0.036	0.054
	WLAN5GHz	802.11n-HT40 MCS0	Back	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.06	0.109	0.164
	WLAN5GHz	802.11n-HT40 MCS0	Left Side	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.07	0.059	0.089
	WLAN5GHz	802.11n-HT40 MCS0	Right Side	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.05	0.140	0.211
	WLAN5GHz	802.11n-HT40 MCS0	Top Side	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.03	0.211	0.317
	WLAN5GHz	802.11n-HT40 MCS0	Bottom Side	0mm	Ant 6	Receiver off	151	5755	8.65	10	1.365	90.72	1.102	0.02	0.001	0.002

【Bluetooth SAR】

Plot No	Band	Mode	Test Position	Gap (mm)	Antenna	Power State	Ch.	Freq. (MHz)	Average Power (dBm)	Tune-Up Limit (dBm)	Tune-Up Scaling Factor	Duty Cycle %	Duty Cycle Scaling Factor	Power Drift (dB)	Measured 10g SAR (W/kg)	Reported 10g SAR (W/kg)
90	Bluetooth	1Mbps	Front	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.06	0.042	0.062
	Bluetooth	1Mbps	Back	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.06	0.061	0.091
	Bluetooth	1Mbps	Left Side	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.14	0.001	0.001
	Bluetooth	1Mbps	Right Side	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	0.18	0.024	0.036
	Bluetooth	1Mbps	Top Side	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.17	0.044	0.065
	Bluetooth	1Mbps	Bottom Side	0mm	Ant 6	Full power	39	2441	8.58	10.00	1.387	77.68	1.072	-0.01	0.001	0.001

<NFC SAR>

Plot No.	Band	Mode	Test Position	Gap (mm)	Freq. (MHz)	Power Drift (dB)	Measured 10g SAR (W/kg)
91	NFC	ASK	Front	0mm	13.56	0.08	0.001
	NFC	ASK	Back	0mm	13.56	0.01	0.011
	NFC	ASK	Left Side	0mm	13.56	0.01	0.001
	NFC	ASK	Right Side	0mm	13.56	0.03	0.001
	NFC	ASK	Top Side	0mm	13.56	-0.08	0.001
	NFC	ASK	Bottom Side	0mm	13.56	-0.04	0.001

15. Simultaneous Transmission Analysis

NO.	Simultaneous Transmission Configurations	MAG1		
		Head	Body	Limbs
1.	WWAN + WLAN2.4GHz	Yes	Yes	Yes
2.	WWAN + WLAN5GHz	Yes	Yes	Yes
3.	WWAN + Bluetooth	Yes	Yes	Yes
4.	WLAN5GHz + Bluetooth	Yes	Yes	Yes
5.	WWAN + WLAN5GHz + Bluetooth	Yes	Yes	Yes
6.	WWAN + WLAN2.4GHz + NFC			Yes
7.	WWAN + WLAN5GHz + NFC			Yes
8.	WWAN + Bluetooth + NFC			Yes
9.	WLAN5GHz + Bluetooth + NFC			Yes
10.	WWAN + WLAN5GHz + Bluetooth + NFC			Yes

Note:

1. This device supports VoIP in GPRS, WCDMA, LTE and 5GNR (e.g. for 3rd-party VoIP) and LTE supports VoLTE operation.
2. The Above table WWAN included 5GNR bands and EN-DC combination.
3. EUT will choose each GSM, WCDMA, LTE and 5GNR according to the network signal condition; therefore, they will not operate simultaneously at any moment.
4. EUT will choose either WLAN 2.4GHz or WLAN 5GHz according to the network signal condition; therefore, 2.4GHz WLAN and 5GHz WLAN will not operate simultaneously at any moment.
5. According to the EUT characteristic, WLAN 5GHz and Bluetooth can transmit simultaneously.
6. According to the EUT characteristic, WLAN 5GHz and WLAN 2.4GHz cannot transmit simultaneously.
7. WLAN 2.4GHz and Bluetooth share the same antenna, and they cannot transmit simultaneously.
8. The NFC only operates on limbs exposure condition.
9. The worst case 5 GHz WLAN SAR for each configuration was used for SAR summation.
10. When stand-alone SAR is not required for a transmitter or antenna, its SAR is considered zero in the SAR summing process to assess Multi-band transmission SAR compliance.
11. For standalone WWAN, always choose the highest SAR among all WWAN bands within the selected antenna for each exposure position to perform simultaneous transmission analysis with WLAN/BT. This is the worst co-located analysis and can represent each band.
12. For EN-DC SAR co-located with WLAN/Bluetooth, chose the worst SAR among the selected LTE bands Bands within the selected antenna per each test position and also the worst SAR of the selected 5GNR Bands within the selected antenna to do co-located with WLAN/Bluetooth. This is the worst co-located analysis and can represent each LTE bands and each 5GNR bands.
13. If 10g-SAR scalar summation < 2.0W/kg for head/Body SAR, < 4.0W/kg for limbs, simultaneous SAR measurement is not necessary.
14. If the 10g-SAR summation > 2.0W/kg for head/body, > 4.0W/kg for limbs, the highest summation SAR is according to EN 62209-2 section6.3.2.4, the procedure is used existing area scans in combination with interpolation and extrapolation for generation of volumetric SAR data by "SMCAD X" software and is a rapid way of obtaining the multiband SAR results, more detail please refers section 15.4.

15.1 Head Exposure Conditions

WWAN1		Exposure Position	1	2	3	4	1+2 Summed 10g SAR (W/kg)	1+3+4 Summed 10g SAR (W/kg)
			WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6		
			10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)		
All	Ant0	Right Cheek	0.176	0.367	0.069	0.001	0.543	0.246
		Right Tilted	0.09	0.288	0.081	0.001	0.378	0.172
		Left Cheek	0.182	0.831	0.105	0.045	1.013	0.332
		Left Tilted	0.101	0.567	0.120	0.001	0.668	0.222
All	Ant4	Right Cheek	0.351	0.367	0.069	0.001	0.718	0.421
		Right Tilted	0.298	0.288	0.081	0.001	0.586	0.380
		Left Cheek	1.052	0.831	0.105	0.045	1.882	1.202
		Left Tilted	0.74	0.567	0.120	0.001	1.307	0.861

<ENDC>

WWAN1		WWAN2		Exposure Position	1	2	3	4	5	1+2+3 Summed 10g SAR (W/kg)	1+2+4+5 Summed 10g SAR (W/kg)
					WWAN1	WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6		
					10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)		
LTE Band 1 LTE Band 3 LTE Band 5 LTE Band 8 LTE Band 41	Ant 0	FR1 n77 FR1 n79	Ant 4	Right Cheek	0.088	0.205	0.367	0.069	0.001	0.660	0.363
				Right Tilted	0.054	0.21	0.288	0.081	0.001	0.552	0.346
				Left Cheek	0.1	0.682	0.831	0.105	0.045	1.613	0.932
				Left Tilted	0.058	0.45	0.567	0.120	0.001	1.075	0.629

15.2 Body Exposure Conditions

WWAN1		Exposure Position	1	2	3	4	1+2 Summed 10g SAR (W/kg)	1+3+4 Summed 10g SAR (W/kg)
			WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6		
			10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)		
All	Ant0	Front	0.738	0.324	0.075	0.001	1.062	0.814
		Back	1.137	0.7	0.223	0.061	1.837	1.421
		Left Side	0.333	0.001	0.026	0.001	0.334	0.36
		Right Side	0.241	0.249	0.131	0.001	0.49	0.373
		Top Side	0.032	0.229	0.264	0.001	0.261	0.297
		Bottom Side	1.482	0.001	0.002	0.001	1.483	1.485
All	Ant4	Front	0.45	0.324	0.075	0.001	0.774	0.526
		Back	1.28	0.7	0.223	0.061	1.98	1.564
		Left Side	0.023	0.001	0.026	0.001	0.024	0.05
		Right Side	1.048	0.249	0.131	0.001	1.297	1.18
		Top Side	0.361	0.229	0.264	0.001	0.59	0.626
		Bottom Side	0.002	0.001	0.002	0.001	0.003	0.005

<ENDC>

WWAN1		WWAN2		Exposure Position	1	2	3	4	5	1+2+3 Summed 10g SAR (W/kg)	1+2+4+5 Summed 10g SAR (W/kg)	Combined
					WWAN1	WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6			
					10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)			
LTE Band 1	Ant 0	FR1 n77	Ant 4	Front	0.532	0.45	0.324	0.075	0.001	1.306	1.058	
				Back	0.871	1.242	0.7	0.223	0.061	2.813	2.397	1&2
				Left Side	0.223	0.002	0.001	0.026	0.001	0.226	0.252	
				Right Side	0.074	0.843	0.249	0.131	0.001	1.166	1.049	
				Top Side	0.001	0.361	0.229	0.264	0.001	0.591	0.627	
				Bottom Side	1.3	0.002	0.001	0.002	0.001	1.303	1.305	
LTE Band 1	Ant 0	FR1 n79	Ant 4	Front	0.532	0.241	0.324	0.075	0.001	1.097	0.849	
				Back	0.871	0.367	0.7	0.223	0.061	1.938	1.522	
				Left Side	0.223	0.002	0.001	0.026	0.001	0.226	0.252	
				Right Side	0.074	1.048	0.249	0.131	0.001	1.371	1.254	
				Top Side	0.001	0.238	0.229	0.264	0.001	0.468	0.504	
				Bottom Side	1.3	0.002	0.001	0.002	0.001	1.303	1.305	
LTE Band 3	Ant 0	FR1 n77	Ant 4	Front	0.407	0.45	0.324	0.075	0.001	1.181	0.933	
				Back	0.624	1.242	0.7	0.223	0.061	2.566	2.15	3&4
				Left Side	0.099	0.002	0.001	0.026	0.001	0.102	0.128	
				Right Side	0.048	0.843	0.249	0.131	0.001	1.14	1.023	
				Top Side	0.001	0.361	0.229	0.264	0.001	0.591	0.627	
				Bottom Side	0.864	0.002	0.001	0.002	0.001	0.867	0.869	
LTE Band 3	Ant 0	FR1 n79	Ant 4	Front	0.407	0.241	0.324	0.075	0.001	0.972	0.724	
				Back	0.624	0.367	0.7	0.223	0.061	1.691	1.275	
				Left Side	0.099	0.002	0.001	0.026	0.001	0.102	0.128	
				Right Side	0.048	1.048	0.249	0.131	0.001	1.345	1.228	
				Top Side	0.001	0.238	0.229	0.264	0.001	0.468	0.504	
				Bottom Side	0.864	0.002	0.001	0.002	0.001	0.867	0.869	
LTE Band 41	Ant 0	FR1 n77	Ant 4	Front	0.244	0.45	0.324	0.075	0.001	1.018	0.77	
				Back	0.283	1.242	0.7	0.223	0.061	2.225	1.809	5
				Left Side	0.135	0.002	0.001	0.026	0.001	0.138	0.164	
				Right Side	0.033	0.843	0.249	0.131	0.001	1.125	1.008	
				Top Side	0.001	0.361	0.229	0.264	0.001	0.591	0.627	
				Bottom Side	0.234	0.002	0.001	0.002	0.001	0.237	0.239	

15.3 Limbs Exposure Conditions

WWAN1		Exposure Position	1	2	3	4	5	1+2+5 Summed 10g SAR (W/kg)	1+3+4+5 Summed 10g SAR (W/kg)	Combined
			WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6	NFC			
			10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)			
All	Ant 0	Front	1.731	1.291	0.192	0.062	0.001	3.023	1.986	
		Back	3.148	1.664	0.433	0.091	0.011	4.823	3.683	6
		Left Side	0.965	0.003	0.089	0.001	0.001	0.969	1.056	
		Right Side	1.147	0.813	0.306	0.036	0.001	1.961	1.49	
		Top Side	0.047	1.164	0.73	0.065	0.001	1.212	0.843	
		Bottom Side	3.133	0.003	0.002	0.001	0.001	3.137	3.137	
All	Ant 4	Front	1.517	1.291	0.192	0.062	0.001	2.809	1.772	
		Back	3.108	1.664	0.433	0.091	0.011	4.783	3.643	7
		Left Side	0.089	0.003	0.089	0.001	0.001	0.093	0.18	
		Right Side	3.125	0.813	0.306	0.036	0.001	3.939	3.468	
		Top Side	1.582	1.164	0.73	0.065	0.001	2.747	2.378	
		Bottom Side	0.068	0.003	0.002	0.001	0.001	0.072	0.072	

<ENDC>

WWAN1		WWAN2		Exposure Position	1	2	3	4	5	6	1+2+3+6 Summed 10g SAR (W/kg)	1+2+4+5+6 Summed 10g SAR (W/kg)	Combined
					WWAN1	WWAN1	2.4GHz WLAN Ant 6	5GHz WLAN Ant 6	Bluetooth Ant 6	NFC			
					10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)	10g SAR (W/kg)			
LTE Band 1	Ant 0	FR1 n77	Ant 4	Front	1.473	1.301	1.291	0.192	0.062	0.001	4.066	3.029	8
				Back	2.38	1.946	1.664	0.433	0.109	0.011	6.001	4.879	9&10
				Left Side	0.506	0.043	0.003	0.089	0.001	0.001	0.553	0.64	
				Right Side	0.13	2.825	0.813	0.306	0.036	0.001	3.769	3.298	
				Top Side	0.042	1.582	1.164	0.73	0.065	0.001	2.789	2.42	
				Bottom Side	2.667	0.068	0.003	0.002	0.001	0.001	2.739	2.739	
LTE Band 1	Ant 0	FR1 n79	Ant 4	Front	1.473	0.329	1.291	0.192	0.062	0.001	3.094	2.057	
				Back	2.38	0.585	1.664	0.433	0.109	0.011	4.64	3.518	11
				Left Side	0.506	0.001	0.003	0.089	0.001	0.001	0.511	0.598	
				Right Side	0.13	3.125	0.813	0.306	0.036	0.001	4.069	3.598	12
				Top Side	0.042	0.294	1.164	0.73	0.065	0.001	1.501	1.132	
				Bottom Side	2.667	0.001	0.003	0.002	0.001	0.001	2.672	2.672	
LTE Band 3	Ant 0	FR1 n77	Ant 4	Front	1.619	1.301	1.291	0.192	0.062	0.001	4.212	3.175	13
				Back	2.715	1.946	1.664	0.433	0.109	0.011	6.336	5.214	14&15
				Left Side	0.519	0.043	0.003	0.089	0.001	0.001	0.566	0.653	
				Right Side	0.08	2.825	0.813	0.306	0.036	0.001	3.719	3.248	
				Top Side	0.027	1.582	1.164	0.73	0.065	0.001	2.774	2.405	
				Bottom Side	1.941	0.068	0.003	0.002	0.001	0.001	2.013	2.013	
LTE Band 3	Ant 0	FR1 n79	Ant 4	Front	1.619	0.329	1.291	0.192	0.062	0.001	3.24	2.203	
				Back	2.715	0.585	1.664	0.433	0.109	0.011	4.975	3.853	16
				Left Side	0.519	0.001	0.003	0.089	0.001	0.001	0.524	0.611	
				Right Side	0.08	3.125	0.813	0.306	0.036	0.001	4.019	3.548	17
				Top Side	0.027	0.294	1.164	0.73	0.065	0.001	1.486	1.117	
				Bottom Side	1.941	0.001	0.003	0.002	0.001	0.001	1.946	1.946	
LTE Band 41	Ant 0	FR1 n77	Ant 4	Front	0.958	1.301	1.291	0.192	0.062	0.001	3.551	2.514	
				Back	1.984	1.946	1.664	0.433	0.109	0.011	5.605	4.483	18&19
				Left Side	0.754	0.043	0.003	0.089	0.001	0.001	0.801	0.888	
				Right Side	0.109	2.825	0.813	0.306	0.036	0.001	3.748	3.277	
				Top Side	0.001	1.582	1.164	0.73	0.065	0.001	2.748	2.379	
				Bottom Side	1.125	0.068	0.003	0.002	0.001	0.001	1.197	1.197	

Note: For the results marked in yellow in the tables above, since the NFC emission transmission SAR scalar sum of the limbs is > 4.0W/kg, therefore, the WWAN and WLAN/BT (except NFC) SAR results are combined using the procedure in Section 6.3.2.4 of IEC 62209-2 and can be referred to section 15.4. And transmitted simultaneously with NFC to show that the SAR scalar sum meets the regulatory requirements.

1	2	1+2
Combined	NFC	Summed
10g SAR (W/kg)	10g SAR (W/kg)	Combined SAR
3.710	0.011	3.721

15.4 Multi-Band Combined SAR Result

General Note:

- During multi-band SAR combination, the duty cycle scaling factors were input into the SEMCAD X software to account for tune-up tolerance.
- The combine SAR result is in the appendix B.

<Body>

Case No	Band	BW (MHz)	Modulation	RB Size	RB offset	Mode	Test Position	Gap (mm)	Antenna	Ch.	Freq. (MHz)	Reported 10g SAR (W/kg)	Summed 10g SAR (W/kg)	Multi-Band Combined 10g SAR (W/kg)
1	LTE Band 1	20M	QPSK	1	0		Back	5mm	Ant 0	18300	1950	0.871	2.813	1.42
	FR1 n77_HPUE	100M	BPSK	135	67		Back	5mm	Ant 4	650000	3750	1.242		
	WLAN2.4GHz					802.11b 1Mbps	Back	5mm	Ant 6	7	2442	0.7		
2	LTE Band 1	20M	QPSK	1	0		Back	5mm	Ant 0	18300	1950	0.871	2.397	1.31
	FR1 n77_HPUE	100M	BPSK	135	67		Back	5mm	Ant 4	650000	3750	1.242		
	WLAN5GHz					802.11n-HT20 MCS0	Back	5mm	Ant 6	140	5700	0.223		
	Bluetooth					1Mbps	Back	5mm	Ant 6	39	2441	0.061		
3	LTE Band 3	20M	QPSK	1	0		Back	5mm	Ant 0	19575	1747.5	0.624	2.566	1.41
	FR1 n77_HPUE	100M	BPSK	135	67		Back	5mm	Ant 4	650000	3750	1.242		
	WLAN2.4GHz					802.11b 1Mbps	Back	5mm	Ant 6	7	2442	0.7		
4	LTE Band 3	20M	QPSK	1	0		Back	5mm	Ant 0	19575	1747.5	0.624	2.150	1.26
	FR1 n77_HPUE	100M	BPSK	135	67		Back	5mm	Ant 4	650000	3750	1.242		
	WLAN5GHz					802.11n-HT20 MCS0	Back	5mm	Ant 6	140	5700	0.223		
	Bluetooth					1Mbps	Back	5mm	Ant 6	39	2441	0.061		
5	LTE Band 41	20M	QPSK	1	0		Back	5mm	Ant 0	40620	2593	0.283	2.225	1.39
	FR1 n77_HPUE	100M	BPSK	135	67		Back	5mm	Ant 4	650000	3750	1.242		
	WLAN2.4GHz					802.11b 1Mbps	Back	5mm	Ant 6	7	2442	0.7		

<Limbs>

Case No	Band	BW (MHz)	Modulation	RB Size	RB offset	Mode	Test Position	Gap (mm)	Antenna	Ch.	Freq. (MHz)	Reported 10g SAR (W/kg)	Summed 10g SAR (W/kg)	Multi-Band Combined 10g SAR (W/kg)
6	LTE Band 7	20M	QPSK	1	0		Back	0mm	Ant 0	20850	2510	3.148	4.812	3.16
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
7	FR1 n41_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	528000	2640	3.108	4.772	3.41
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
8	LTE Band 1	20M	QPSK	1	0		Front	0mm	Ant 0	18300	1950	1.473	4.065	1.98
	FR1 n77_HPUE	100M	BPSK	135	67		Front	0mm	Ant 4	650000	3750	1.301		
	WLAN2.4GHz					802.11b 1Mbps	Front	0mm	Ant 6	7	2442	1.291		
9	LTE Band 1	20M	QPSK	1	0		Back	0mm	Ant 0	18500	1970	2.38	5.990	2.73
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
10	LTE Band 1	20M	QPSK	1	0		Back	0mm	Ant 0	18500	1970	2.38	4.868	2.39
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN5GHz					802.11n-HT20 MCS0	Back	0mm	Ant 6	140	5700	0.433		
	Bluetooth					1Mbps	Back	0mm	Ant 6	39	2441	0.109		
11	LTE Band 1	20M	QPSK	1	0		Back	0mm	Ant 0	18500	1970	2.38	4.629	2.39

	FR1 n79_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	730000	4950	0.585		
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
12	LTE Band 1	20M	QPSK	1	0		Right Side	0mm	Ant 0	18300	1950	0.13	4.068	3.71
	FR1 n79_HPUE	100M	BPSK	135	67		Right Side	0mm	Ant 4	730000	4950	3.125		
	WLAN2.4GHz					802.11b 1Mbps	Right Side	0mm	Ant 6	7	2442	0.813		
13	LTE Band 3	20M	QPSK	1	0		Front	0mm	Ant 0	19575	1747.5	1.619	4.211	1.98
	FR1 n77_HPUE	100M	BPSK	135	67		Front	0mm	Ant 4	650000	3750	1.301		
	WLAN2.4GHz					802.11b 1Mbps	Front	0mm	Ant 6	7	2442	1.291		
14	LTE Band 3	20M	QPSK	1	0		Back	0mm	Ant 0	19300	1720	2.715	6.325	2.73
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
15	LTE Band 3	20M	QPSK	1	0		Back	0mm	Ant 0	19300	1720	2.715	5.203	2.82
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN5GHz					802.11n-HT20 MCS0	Back	0mm	Ant 6	140	5700	0.433		
	Bluetooth					1Mbps	Back	0mm	Ant 6	39	2441	0.109		
16	LTE Band 3	20M	QPSK	1	0		Back	0mm	Ant 0	19300	1720	2.715	4.964	2.72
	FR1 n79_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	730000	4950	0.585		
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
17	LTE Band 3	20M	QPSK	1	0		Right Side	0mm	Ant 0	19575	1747.5	0.08	4.018	3.71
	FR1 n79_HPUE	100M	BPSK	135	67		Right Side	0mm	Ant 4	730000	4950	3.125		
	WLAN2.4GHz					802.11b 1Mbps	Right Side	0mm	Ant 6	7	2442	0.813		
18	LTE Band 41	20M	QPSK	1	0		Back	0mm	Ant 0	40620	2593	1.984	5.594	2.61
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN2.4GHz					802.11b 1Mbps	Back	0mm	Ant 6	7	2442	1.664		
19	LTE Band 41	20M	QPSK	1	0		Back	0mm	Ant 0	40620	2593	1.984	4.472	2.04
	FR1 n77_HPUE	100M	BPSK	135	67		Back	0mm	Ant 4	650000	3750	1.946		
	WLAN5GHz					802.11n-HT20 MCS0	Back	0mm	Ant 6	140	5700	0.433		
	Bluetooth					1Mbps	Back	0mm	Ant 6	39	2441	0.109		

Test Engineer : Martin Li, Varus Wang, Light Wang, Shaochen Kang, Chenchen Yang

16. Uncertainty Assessment

The component of uncertainty may generally be categorized according to the methods used to evaluate them. The evaluation of uncertainty by the statistical analysis of a series of observations is termed a Type A evaluation of uncertainty. The evaluation of uncertainty by means other than the statistical analysis of a series of observation is termed a Type B evaluation of uncertainty. Each component of uncertainty, however evaluated, is represented by an estimated standard deviation, termed standard uncertainty, which is determined by the positive square root of the estimated variance.

A Type A evaluation of standard uncertainty may be based on any valid statistical method for treating data. This includes calculating the standard deviation of the mean of a series of independent observations; using the method of least squares to fit a curve to the data in order to estimate the parameter of the curve and their standard deviations; or carrying out an analysis of variance in order to identify and quantify random effects in certain kinds of measurement.

A type B evaluation of standard uncertainty is typically based on scientific judgment using all of the relevant information available. These may include previous measurement data, experience, and knowledge of the behavior and properties of relevant materials and instruments, manufacture's specification, data provided in calibration reports and uncertainties assigned to reference data taken from handbooks. Broadly speaking, the uncertainty is either obtained from an outdoor source or obtained from an assumed distribution, such as the normal distribution, rectangular or triangular distributions indicated in Table below

Uncertainty Distributions	Normal	Rectangular	Triangular	U-Shape
Multi-plying Factor ^(a)	1/ κ ^(b)	1/ $\sqrt{3}$	1/ $\sqrt{6}$	1/ $\sqrt{2}$

(a) standard uncertainty is determined as the product of the multiplying factor and the estimated range of variations in the measured quantity

(b) κ is the coverage factor

Standard Uncertainty for Assumed Distribution

The combined standard uncertainty of the measurement result represents the estimated standard deviation of the result. It is obtained by combining the individual standard uncertainties of both Type A and Type B evaluation using the usual "root-sum-squares" (RSS) methods of combining standard deviations by taking the positive square root of the estimated variances.

Expanded uncertainty is a measure of uncertainty that defines an interval about the measurement result within which the measured value is confidently believed to lie. It is obtained by multiplying the combined standard uncertainty by a coverage factor. Typically, the coverage factor ranges from 2 to 3. Using a coverage factor allows the true value of a measured quantity to be specified with a defined probability within the specified uncertainty range. For purpose of this document, a coverage factor two is used, which corresponds to confidence interval of about 95 %. The DASY uncertainty Budget is shown in the following tables.

Error Description	Uncertainty Value (±%)	Probability	Divisor	(Ci) 1g	(Ci) 10g	Standard Uncertainty (1g) (±%)	Standard Uncertainty (10g) (±%)
Measurement System							
Probe Calibration	6.00	N	1	1	1	6.0	6.0
Axial Isotropy	4.70	R	1.732	0.7	0.7	1.9	1.9
Hemispherical Isotropy	9.60	R	1.732	0.7	0.7	3.9	3.9
Boundary Effects	1.00	R	1.732	1	1	0.6	0.6
Linearity	4.70	R	1.732	1	1	2.7	2.7
System Detection Limits	1.00	R	1.732	1	1	0.6	0.6
Modulation Response	3.50	R	1.732	1	1	2.0	2.0
Readout Electronics	0.30	N	1	1	1	0.3	0.3
Response Time	0.00	R	1.732	1	1	0.0	0.0
Integration Time	2.60	R	1.732	1	1	1.5	1.5
RF Ambient Noise	3.00	R	1.732	1	1	1.7	1.7
RF Ambient Reflections	3.00	R	1.732	1	1	1.7	1.7
Probe Positioner	0.40	R	1.732	1	1	0.2	0.2
Probe Positioning	2.90	R	1.732	1	1	1.7	1.7
Max. SAR Eval.	2.00	R	1.732	1	1	1.2	1.2
Test Sample Related							
Device Positioning	3.04	N	1	1	1	3.0	3.0
Device Holder	3.43	N	1	1	1	3.4	3.4
Power Drift	5.00	R	1.732	1	1	2.9	2.9
Power Scaling	3.04	R	1.732	1	1	0.0	0.0
Phantom and Setup							
Phantom Uncertainty	6.10	R	1.732	1	1	3.5	3.5
SAR correction	0.11	R	1.732	1	0.84	0.0	0.0
Liquid Conductivity Repeatability	5.00	N	1	0.78	0.71	0.1	0.1
Liquid Conductivity (target)	2.50	R	1.732	0.78	0.71	2.3	2.0
Liquid Conductivity (mea.)	0.10	R	1.732	0.78	0.71	1.1	1.0
Temp. unc. - Conductivity	5.00	R	1.732	0.78	0.71	1.4	1.2
Liquid Permittivity Repeatability	2.50	N	1	0.23	0.26	0.0	0.0
Liquid Permittivity (target)	6.10	R	1.732	0.23	0.26	0.7	0.8
Liquid Permittivity (mea.)	0.11	R	1.732	0.23	0.26	0.3	0.4
Temp. unc. - Permittivity	0.59	R	1.732	0.23	0.26	0.1	0.1
Combined Std. Uncertainty						11.4%	11.3%
Coverage Factor for 95 %						K=2	K=2
Expanded STD Uncertainty						22.8%	22.7%

Uncertainty Budget for frequency range 300 MHz to 3 GHz according to EN 62209-1

Error Description	Uncertainty Value (±%)	Probability	Divisor	(Ci) 1g	(Ci) 10g	Standard Uncertainty (1g) (±%)	Standard Uncertainty (10g) (±%)
Measurement System							
Probe Calibration	7.00	N	1	1	1	7.0	7.0
Axial Isotropy	4.70	R	1.732	0.7	0.7	1.9	1.9
Hemispherical Isotropy	9.60	R	1.732	0.7	0.7	3.9	3.9
Boundary Effects	2.00	R	1.732	1	1	1.2	1.2
Linearity	4.70	R	1.732	1	1	2.7	2.7
System Detection Limits	1.00	R	1.732	1	1	0.6	0.6
Modulation Response	3.46	R	1.732	1	1	2.0	2.0
Readout Electronics	0.30	N	1	1	1	0.3	0.3
Response Time	0.00	R	1.732	1	1	0.0	0.0
Integration Time	2.60	R	1.732	1	1	1.5	1.5
RF Ambient Noise	3.00	R	1.732	1	1	1.7	1.7
RF Ambient Reflections	3.00	R	1.732	1	1	1.7	1.7
Probe Positioner	0.40	R	1.732	1	1	0.2	0.2
Probe Positioning	6.70	R	1.732	1	1	3.9	3.9
Max. SAR Eval.	4.00	R	1.732	1	1	2.3	2.3
Test Sample Related							
Device Positioning	3.04	N	1	1	1	3.0	3.0
Device Holder	3.43	N	1	1	1	3.4	3.4
Power Drift	5.00	R	1.732	1	1	2.9	2.9
Power Scaling	0.00	R	1.732	1	1	0.0	0.0
Phantom and Setup							
Phantom Uncertainty	6.60	R	1.732	1	1	3.8	3.8
SAR correction	0.00	R	1.732	1	0.84	0.0	0.0
Liquid Conductivity Repeatability	0.11	N	1	0.78	0.71	0.1	0.1
Liquid Conductivity (target)	5.00	R	1.732	0.78	0.71	2.3	2.0
Liquid Conductivity (mea.)	2.50	R	1.732	0.78	0.71	1.1	1.0
Temp. unc. - Conductivity	3.01	R	1.732	0.78	0.71	1.4	1.2
Liquid Permittivity Repeatability	0.10	N	1	0.23	0.26	0.0	0.0
Liquid Permittivity (target)	5.00	R	1.732	0.23	0.26	0.7	0.8
Liquid Permittivity (mea.)	2.50	R	1.732	0.23	0.26	0.3	0.4
Temp. unc. - Permittivity	0.59	R	1.732	0.23	0.26	0.1	0.1
Combined Std. Uncertainty						12.7%	12.7%
Coverage Factor for 95 %						K=2	K=2
Expanded STD Uncertainty						25.5%	25.4%

Uncertainty Budget for frequency range 3 GHz to 6 GHz according to EN 62209-1

Error Description	Uncertainty Value (±%)	Probability	Divisor	(Ci) 1g	(Ci) 10g	Standard Uncertainty (1g) (±%)	Standard Uncertainty (10g) (±%)
Measurement System							
Probe Calibration	7.00	N	1	1	1	7.0	7.0
Axial Isotropy	4.70	R	1.732	0.7	0.7	1.9	1.9
Hemispherical Isotropy	9.60	R	1.732	0.7	0.7	3.9	3.9
Linearity	4.70	R	1.732	1	1	2.7	2.7
Modulation Response	3.46	R	1.732	1	1	2.0	2.0
System Detection Limits	1.00	R	1.732	1	1	0.6	0.6
Boundary Effects	2.00	R	1.732	1	1	1.2	1.2
Readout Electronics	0.30	N	1	1	1	0.3	0.3
Response Time	0.00	R	1.732	1	1	0.0	0.0
Integration Time	2.60	R	1.732	1	1	1.5	1.5
RF Ambient Noise	3.00	R	1.732	1	1	1.7	1.7
RF Ambient Reflections	3.00	R	1.732	1	1	1.7	1.7
Probe Positioner	0.40	R	1.732	1	1	0.2	0.2
Probe Positioning	6.70	R	1.732	1	1	3.9	3.9
Post-processing	4.00	R	1.732	1	1	2.3	2.3
Test Sample Related							
Device Holder	3.43	N	1	1	1	3.4	3.4
Test sample Positioning	3.04	N	1	1	1	3.0	3.0
Power Scaling	0.00	R	1.732	1	1	0.0	0.0
Power Drift	5.00	R	1.732	1	1	2.9	2.9
Phantom and Setup							
Phantom Uncertainty	7.60	R	1.732	1	1	4.4	4.4
SAR correction	0.00	R	1.732	1	0.84	0.0	0.0
Liquid Conductivity Repeatability	0.11	N	1	0.78	0.71	0.1	0.1
Liquid Conductivity (target)	5.00	R	1.732	0.78	0.71	2.3	2.0
Liquid Conductivity (mea.)	2.50	R	1.732	0.78	0.71	1.1	1.0
Temp. unc. - Conductivity	3.01	R	1.732	0.78	0.71	1.4	1.2
Liquid Permittivity Repeatability	0.10	N	1	0.23	0.26	0.0	0.0
Liquid Permittivity (target)	5.00	R	1.732	0.23	0.26	0.7	0.8
Liquid Permittivity (mea.)	2.50	R	1.732	0.23	0.26	0.3	0.4
Temp. unc. - Permittivity	0.59	R	1.732	0.23	0.26	0.1	0.1
Combined Std. Uncertainty						12.9%	12.7%
Coverage Factor for 95 %						K=2	K=2
Expanded STD Uncertainty						25.8%	25.4%

Uncertainty Budget for frequency range 30 MHz to 6 GHz according to EN 62209-2

Uncertainty Budget According to EN IEC/IEEE 62209-1528 (Frequency band: 4 MHz - 10 GHz range)							
Error Description	Uncert. Value (±%)	Prob. Dist.	Div.	(Ci) 1g	(Ci) 10g	Standard Uncertainty (1g) (±%)	Standard Uncertainty (10g) (±%)
Measurement System errors							
Probe calibration	18.6	N	2	1	1	9.3	9.3
Probe calibration drift	1.7	R	1.732	1	1	1.0	1.0
Probe linearity and detection Limit	4.7	R	1.732	1	1	2.7	2.7
Broadband signal	2.8	R	1.732	1	1	1.6	1.6
Probe isotropy	7.6	R	1.732	1	1	4.4	4.4
Other probe and data acquisition errors	2.4	N	1	1	1	2.4	2.4
RF ambient and noise	1.8	N	1	1	1	1.8	1.8
Probe positioning errors	0.006	N	1	0.5	0.5	0.0	0.0
Data processing errors	4.0	N	1	1	1	4.0	4.0
Phantom and Device Errors							
Measurement of phantom conductivity (σ)	2.5	N	1	0.78	0.71	2.0	1.8
Temperature effects (medium)	5.4	R	1.732	0.78	0.71	2.4	2.2
Shell permittivity	14.0	R	1.732	0.5	0.5	4.0	4.0
Distance between the radiating element of the DUT and the phantom medium	2.0	N	1	2	2	4.0	4.0
Repeatability of positioning the DUT or source against the phantom	1.0	N	1	1	1	1.0	1.0
Device holder effects	3.6	N	1	1	1	3.6	3.6
Effect of operating mode on probe sensitivity	2.4	R	1.732	1	1	1.4	1.4
Time-average SAR	1.7	R	1.732	1	1	1.0	1.0
Variation in SAR due to drift in output of DUT	2.5	N	1	1	1	2.5	2.5
Validation antenna uncertainty (validation measurement only)	0.0	N	1	1	1	0.0	0.0
Uncertainty in accepted power (validation measurement only)	0.0	N	1	1	1	0.0	0.0
Correction to the SAR results							
Phantom deviation from target (ϵ' , σ)	1.9	N	1	1	0.84	1.9	1.6
SAR scaling	0.0	R	1.732	1	1	0.0	0.0
Combined Std. Uncertainty						14.5%	14.4%
Coverage Factor for 95 %						K=2	K=2
Expanded STD Uncertainty						29.0%	28.8%

17. References

- [1] Council Recommendation 1999/519/EC of July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz)
- [2] EN 50360: 2017, EN 50360:2017/A1:2023, "Product standard to demonstrate the compliance of wireless communication devices, with the basic restrictions and exposure limit values related to human exposure to electromagnetic fields in the frequency range from 300 MHz to 6 GHz: devices used next to the ear."
- [3] EN 50566:2017, EN 50566:2017/A1:2023, "Product standard to demonstrate the compliance of wireless communication devices with the basic restrictions and exposure limit values related to human exposure to electromagnetic fields in the frequency range from 30 MHz to 6 GHz: hand-held and body mounted devices in close proximity to the human body"
- [4] EN IEC 62311:2020, "Assessment of electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz – 300 GHz)"
- [5] EN 50665:2017, "Generic standard for assessment of electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz - 300 GHz)"
- [6] EN 62209-1:2016, "Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices- Devices used next to the ear (Frequency range of 300 MHz to 6 GHz)"
- [7] EN 62209-2:2010, EN 62209-2:2010/A1:2019, "Human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices. Human models, instrumentation, and procedures. Procedure to determine the specific absorption rate (SAR) for wireless communication devices used in close proximity to the human body (frequency range of 30 MHz to 6 GHz)"
- [8] EN 62479:2010 "Assessment of the compliance of low power electronic and electrical equipment with the basic restrictions related to human exposure to electromagnetic fields (10 MHz to 300 GHz)"
- [9] EN 50663:2017 " Generic standard for assessment of low power electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz - 300 GHz)"
- [10] EN IEC/IEEE 62209-1528:2021, "Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices – Part 1528: Human models, instrumentation, and procedures (Frequency range of 4 MHz to 10 GHz)"
- [11] SPEAG DASY System Handbook



Appendixes

Please refer to separated files for the following appendixes

Appendix A. Plots of System Performance Check

Appendix B. Plots of High SAR Measurement

Appendix C. DASY Calibration Certificate

Appendix D. Test Setup Photos

Appendix E. Conducted RF Output Power Table

-----THE END-----